Automation in Construction 137 (2022) 104191

Contents lists available at ScienceDirect

Automation in Construction

journal homepage: www.elsevier.com/locate/autcon

Graph-based linguistic and visual information integration for on-site occupational hazards identification

Shi Chen *, Kazuyuki Demachi, Feiyan Dong

Department of Nuclear Engineering and Management, School of Engineering, The University of Tokyo, Tokyo, Japan

ARTICLE INFO

Keywords:
Construction safety
Occupational hazards identification
Personal protective equipment (PPE)
Deep learning
Graph

ABSTRACT

Construction sites are hazardous with various potential hazards that can occur at any time. The combination of different factors always causes the construction fatalities, and the majority of these fatalities could be prevented if workers followed on-site regulatory rules. However, compliance of regulatory rules is not strictly enforced among workers due to all kinds of reasons. Although previously proposed vision-based approaches are available for occupational hazards identification, the practicality is limited by the lack of automated understanding and adaptability to regulatory rules changes. In response to these gaps, this paper proposes a novel graph-based framework that integrates linguistic and visual information to process regulatory rule sentences and images for on-site occupational hazards identification. Particularly, a regulatory rules processing approach is presented to automatically extract and represent the key linguistic information of regulatory rules and a vision-based image scene information understanding approach is introduced to process on-site images by the combination of deep learning-based object detection and individual detection using geometric relationships analysis. Additionally, an automated reasoning approach is proposed to provide the integration of the processed linguistic and visual information and perform hazards identification. The hazards of two scenes, i.e., "working on height" and "operating a grinder", were successfully identified with significantly higher performance compared to the baseline model.

1. Introduction

The construction industry is a very hazard intensive field, with the construction phase being the one with the most occurrences of occupational accidents. According to the United States' Bureau of Labor Statistics (BLS), the number of construction fatalities in the US has gradually increased from 924 to 1066 between 2015 and 2019 [1]. Similarly, in Japan, there were 1522 construction fatal accidents during 2015–2019 [2]. These accidents not only resulted in irreparable fatalities and economic losses, but also restricted the sustainable development of the construction industry to a certain extent.

As the construction is extremely vulnerable to the interference of various subjective and objective factors, once any sudden problem occurs it will pose a potential threat to the life and property safety of the on-site construction workers. However, the majority of these fatalities could be prevented if workers wore appropriate personal protective equipment (PPE), e.g., hard hats, gloves, body harness [3], and followed on-site regulatory rules. The consequences of head injuries caused by falling from height or being stuck by vehicles and other moving plants and equipment are one of the most serious of all construction accidents. A total of 2210 construction fatalities occurred in the United States because of traumatic brain injury (TBI), which represented 25% of all construction fatalities during 2003 and 2010 [4]. The Occupational Safety and Health Administration (OSHA) in the United States stipulated that workers working in areas where there is a possible danger of head injury from impact, or from falling or flying objects, or from electrical shock and burns shall be protected by hard hats [5]. Construction-related occupational eye injuries are an important cause of vision loss. According to the National Institute for Occupational Safety and Health (NIOSH), an average of 2000 United States workers require medical treatment for job-related eye injuries every day [6]. The majority of construction-related eye injuries are preventable. The reasons cited for the majority of eye injuries include the non-wearing of available eye protection or wearing of inappropriate eye protection for the current task [7]. Fine dust and particles, gases and vapors can be produced when using machine tools. Silica dust from bricks can cause lung and airway diseases such as emphysema, bronchitis, silicosis, and may increase the risk of cancer. PPE, such as respirators or dust masks, are used to

* Corresponding author.
E-mail address: shichen@g.ecc.u-tokyo.ac.jp (S. Chen).

https://doi.org/10.1016/j.autcon.2022.104191
Received 25 March 2021; Received in revised form 9 January 2022; Accepted 26 February 2022
Available online 8 March 2022

controls these hazards [8]. OSHA indicates the workers shall be ensured to wear eye or face protection when exposed to eye or face hazards from flying particles, molten metal, liquid chemicals, acids or caustic liquids, chemical gases or vapors, or potentially injurious light radiation [9]. Construction workers who are six feet or more above lower levels are at risk for serious injury or death if they should fall. According to BLS, there were 320 fatal falls to a lower level out of 1008 construction fatalities in 2018, which indicates fall accidents are a substantial burden and an impediment to accomplishing occupational safety in the construction industry [10].

Nonetheless, even with prior education and training, construction workers may not precisely follow on-site safety regulations due to reasons such as momentary carelessness, fatigue working. Thus, on-site occupational safety monitoring is becoming increasingly important for safety management and an automated on-site occupational hazards identification system which of the effectiveness of designs in reducing human factors error and is able to carry out occupational hazards identification to predict the prevention of all kinds of accidents accurately, needs to be taken into consideration. At present, vision-based occupational hazards identification are gradually being widely used due to the extensive deployment of surveillance cameras on construction sites. However, in order to achieve precise site management, regulatory rules vary from region to region and may change due to external conditions. For example, workers were generally required to wear masks only in areas where large amounts of dust could be generated, but due to the spread of the COVID-19 pandemic, the mask wearing requirement was modified to apply to the entire construction site. Therefore the ability of automatically understanding regulatory rules to dynamically respond to the diversity of on-site safety management requirements will be a major challenge for the practicality of current vision-based occupational hazards identification systems.

In response to these difficulties, this paper proposes a graph-based framework that integrates linguistic and visual information to process regulatory rule sentences and images for on-site occupational hazards identification. As the first step toward the end, an automated regulatory rules processing approach is proposed to extract and represent the key linguistic information of regulatory rules that is intended to indicate the potential occupational hazards. The proposed approach utilizes natural language processing (NLP)-based syntactic structure analysis to automatically generate syntactic features and represent key linguistic information in a computable hierarchical graph structure based on automated ontology modeling. Subsequently, a vision-based image scene information understanding approach is developed to process on-site images. Individuals' body joint positions and objects (e.g., PPE, tools) are detected by the deep learning-based pose estimation model and object detection model, respectively. Relationships between detected individuals and objects are further obtained by geometric relationship analysis to represent the visual information in the scene graph. Finally, an automated reasoning approach based on graph structure analysis is deployed to provide the integration of obtained linguistic and visual information and perform hazards identification. In the experiments, the proposed model was able to identify the hazards of two scenes, i.e., "working on height" and "operating a grinder", with high precision and recall. The experimental results demonstrate the effectiveness and robustness of the proposed approach for on-site occupational hazards identification.

2. Related works

In this section, we first review the traditional vision-based approaches for on-site hazards identification, including approaches that use deep learning-based object detection, and novel approaches that combine deep learning-based object detection with pose estimation. Next, we investigate approaches that integrate linguistic information with vision-based models to address the requirements for improved practicality and robustness. Finally, we summarize the benefits and drawbacks of these approaches.

2.1. Vision-based occupational hazards identification approaches

Vision-based approaches are now widely used for automated identification of occupational hazards, as deep learning-based object detection methods have shown remarkable performance on most visual tasks in the architecture, engineering and construction (AEC) industry. Fang et al. [11] proposed an object detection-based method using Faster R-CNN to automatically detect construction workers' non-hardhat-use (NHU). A total of 81,000 image frames were collected from various construction sites as a training dataset and the bounding boxes that surround workers in the images were annotated as the ground truth to train the model. Wu et al. [12] deployed Single Shot Multibox Detector (SSD) with presented reverse progressive attention (RPA) for NHU identification. A benchmark dataset GDUT-HWD was created by downloading Internet images retrieved by search engines to train the SSD-RPA model. In contrast to [11], only the head regions of workers were annotated as the ground truth. Nath et al. [13] introduced and tested models built on YOLOv3 architecture to verify PPE (hard hat and vest) compliance of workers. Three approaches were verified concerning different classifiers (e.g., decision tree, VGG-16, ResNet-50, Xception, or Bayesian). In their extended work [14], the Pictor-v2 dataset was introduced, which contains 1105 crowd-sourcing and 1402 web-mined annotated images of building, equipment, and worker. This dataset was used to train, validate, and test YOLO algorithms for real-time detection of these three common object categories in construction imagery. Shen et al. [15] proposed a NHU identification approach by making use of DenseNet-based face detection and bounding box regression. These approaches generally consider the bounding box of workers as the output of the object detection model, which is indeed well achieved in single hazard identification tasks (e.g., NHU) with a binary classification, i.e., "individual using PPE" and "individual not using PPE". Nevertheless, when it comes to a multi-hazard identification task, e.g., n classes of appropriate PPE use identification, the output will increase to 2^n, which may lead to difficulties in training data preparation and degradation of detection performance. Recently, deep learning-based pose estimation algorithms have achieved impressive results in unconstrained environments, showing the potential for worker detection in complex on-site environments. Studies have been carried out to compensate for the insufficiency of object detection-based approaches for occupational hazards by combining deep learning-based pose estimation models [16,17]. Chen et al. [16] introduced a vision-based approach to detect the proper use of multi-class radiation PPE in nuclear power plants via a combination of deep learning-based object detection and pose estimation using Euclidean distance between bounding boxes of detected PPE and the neck keypoint of detected individual. Xiong et al. [17] presented an extensible pose-guided anchoring framework aimed at multi-class PPE compliance detection. A pose estimator was deployed to detect individuals and provide joint-level anchors for guiding the localization of different PPE items. Compared to object detection-based approaches, human skeletons provide more fine-grained information about a person (e.g., location and visibility), especially in the case of occlusion, and is more effective for multi-hazard identification task (needs only n outputs for a n-class hazard identification task).

2.2. Integrating linguistic and visual information for occupational hazards identification

As regulatory rules may change at any time in practical engineering, a occupational hazards identification model with the ability to dynamically adapt to rule adjustments would be more practical and robust. To this end, a unified model integrating linguistic and visual information would enable the automatic and effective identification of hazards in compliance with regulatory rules even when changes are made to them.

Several explorations to this end have already been carried out [18–23]. Xiong et al. [18] developed an Automated Hazards Identification System (AHIS) to evaluate the operation descriptions generated from site videos against the safety guidelines extracted from the textual documents with the assistance of the ontology of construction safety. Two types of crucial hazards, i.e., failing to wear a hard hat and walking beneath the cane, were successfully identified. Zhang et al. [19] proposed an automatic identification method that integrating object detection and ontology to identify risks in the construction process and prevent construction accidents. Fang et al. [20] integrated computer vision algorithms with ontology models to develop a knowledge graph that consists of an ontological model for hazards, knowledge extraction, and knowledge inference for hazard identification, which can automatically identify falls from heights hazards in varying contexts from images. Wu et al. [21] introduced a conceptual framework that combines computer vision and ontology techniques to semantically reasoning falls from height (FFH) hazards. Mask R-CNN was used to detect visual information from on-site photos while the safety regulatory knowledge was represented by ontology and semantic web rule language (SWRL) rules. Knan et al. [22] proposed a correlation-based approach for mobile scaffold safety monitoring and detecting worker's unsafe behaviors. The scaffolding related regulations were analyzed manually. Mask R-CNN was used as classification and segmentation of worker's tasks combined with object correlation detection (OCD) module to identify worker's unsafe behaviors. However, both [18–22] require manual processing of regulatory information to establish the taxonomy of hazards, which is a time-consuming and error-prone task. In addition, the implementation of hazards reasoning in these works requires third-party software (e.g., Protégé.) rather than being performed in a unified model, which reduces their practicality. Liu et al. [23] initially established a linguistic description schema for manifesting scenes, and created two unique dedicated image captioning datasets for method validation. A general model architecture for image captioning was then developed by combining an encoder-decoder framework with a deep neural network for manifesting construction activity scenes. However, it does not provide automatic compliance checking of regulatory rules.

2.3. Summary

Notably, literature survey has indicated the following issues and directions that are of great interest both academically and in engineering terms.

(1) The widely used vision-based occupational hazards identification approaches [11–15] using deep learning-based object detection models are not applicable when performing multi-hazard identification tasks. In contrast, a combination of deep learning-based object detection and pose estimation models [16,17] can be more effective for identification.
(2) In order to improve the practicality and robustness of occupational hazards identification models to adapt to dynamic changes in regulatory rules, models that integrate linguistic information representation would be an enhancement to current vision-based models.
(3) Investigations have been carried out to integrate linguistic and visual information to automatically and efficiently identify hazards according to regulatory rules [18–23]. However, these works provided no solution for automated regulatory information processing and the manual process is time-consuming, costly, and error-prone. Besides, their hazard reasoning modules require the use of third-party software. Such factors affect the practicality of these approaches for on-site deployment.

3. Methodology

In order to address the aforementioned issues, the work presented in this paper is dedicated to the proposal and development of a graph-based framework for occupational hazards identification integrating linguistic and visual information with (1) perceptual capabilities that enable automated extraction of descriptive key information from text for regulatory understanding (e.g., *prohibit* using tool *A* to perform operation *B*) and entities (individuals and objects) from on-site images for scene understanding (e.g., a worker is using tool *A* to perform operation *B*), and (2) reasoning capabilities that automatically perform compliance checking of regulatory rules covering improper use of PPE and risky individual behaviors at construction sites. In this section, the proposed framework is described in detail.

3.1. Overview of the proposed framework

The generic pipeline of the proposed framework is illustrated in Fig. 1, which comprises three modules:

(1) **Linguistic information extraction module** automatically processes all the input regulatory rules and transforms natural language sentences into computable graph structure representation through syntactic feature generation and semantic analysis.
(2) **Visual information extraction module** automatically extracts the entities (individuals and objects) in the input image and performs relationship analysis between entities for graph structure representation. In contrast to the approaches of construction sites scene understanding using only object detection models or pose estimation models, the visual information extraction module in this paper combines the two for a more effective representation of scene information. Additionally, the notion of scene graph is used for visual information representation, which is a calculable structure and allows to represent the richness of entities and relationships between entities that can exist in an intuitive graph-structure.
(3) **Automated reasoning module** takes the graphs output from the linguistic and visual information extraction modules as input and performs occupational hazards identification based on graph structure analysis.

The proposed framework in this paper is based on our previous work [24], compared to which this paper has the following improvements:

(1) This paper originally proposes an automated linguistic information extraction approach using NLP-based syntactic structure analysis with ontology modeling to extract key information from regulatory rules. It is more practical and efficient than the manual regulatory information extraction used in [24]. The extracted linguistic information is represented by a novel hierarchical graph structure proposed in [24], which is further refined in this paper by addressing the prohibition relationships in regulatory rules.
(2) In contrast to [24], this paper employs the existing state-of-the-art vision-based object detection model, i.e., YOLOv4, resulting in better scene understanding performance compared to [24].
(3) Apart from the proper use identification of PPE addressed in [24], this paper also performs the automatic identification of the safe operation of grinder, which is a relatively more challenging task.

3.2. Linguistic information extraction

The manual process of regulatory rules is time-consuming, costly and error-prone due to the large number of regulatory documents, the differences in format and semantics of their provisions, and the volume and complexity of the information described. To address this gap, we propose a semantic analysis approach based on NLP and ontology to extract linguistic information for automated regulatory rule processing.

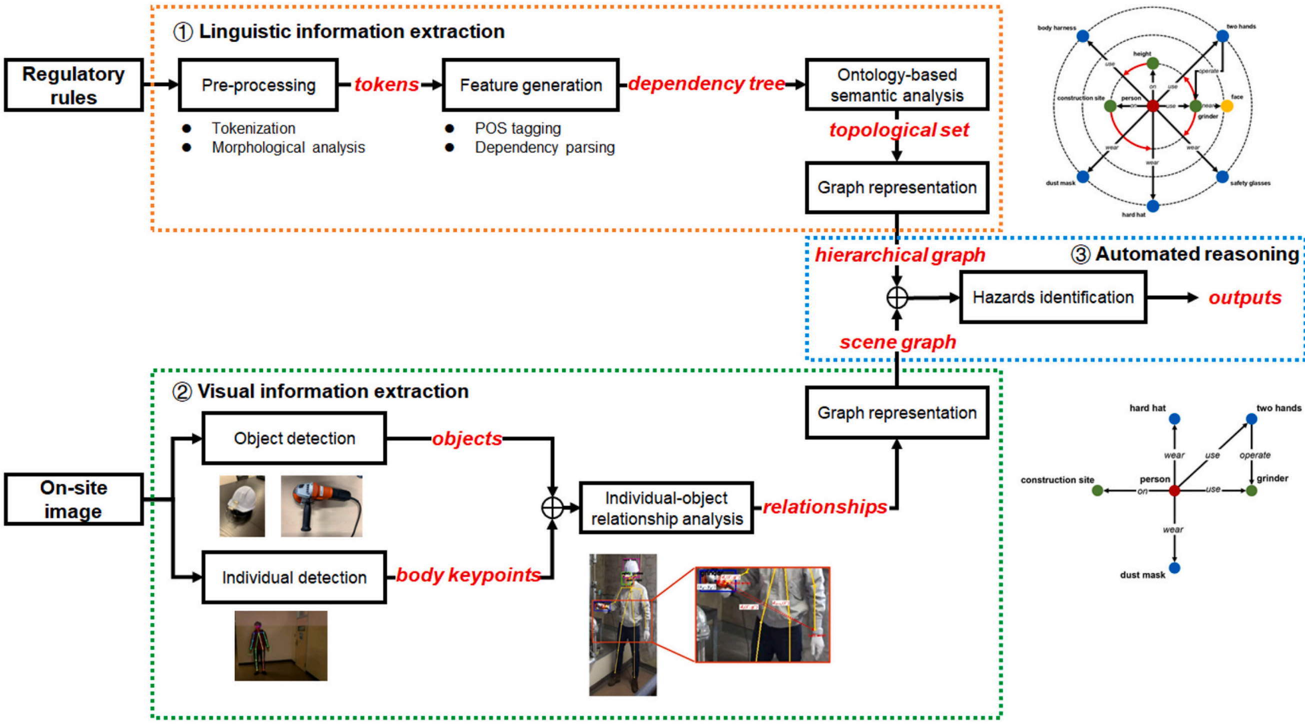


Fig. 1. Generic pipeline of the proposed framework.

3.2.1. Pre-processing

Regulatory rules are first pre-processed to prepare the raw text for further analysis, which consists of tokenization and morphological analysis.

(1) Tokenization is the process of splitting a phrase, sentence, paragraph, or an entire text document into smaller units, such as sentences or words. Each of these smaller units is called tokens [25]. The regulatory rules are divided into tokens, where a token is a single word, a number, a punctuation mark, a whitespace characters, or a symbol. This process is intended to prepare the text for further unit-based processing, which parses the text according to common delimiters (i.e., white spaces and punctuations) with disambiguation consideration (e.g., "," as a delimiter in a number instead of punctuation). As shown in Fig. 2, given the raw text from regulatory documents, boundaries of the token are recognized and labeled out using the "<token>" (i.e., starting of a token) or "</token>" (i.e., ending of a token) tags.

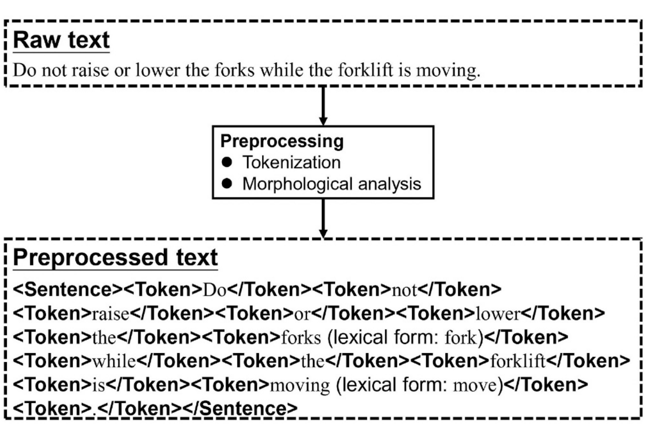


Fig. 2. An example of text preprocessing.

(2) Morphological analysis is performed to map various nonstandard forms of a word (e.g., the plural form of a noun, the past tense of a verb) to its lexical form (e.g., the singular form of a noun, the infinitive form of a verb). As an example, in Fig. 2, "forks" and "moving" are mapped to their lexical forms "fork" and "move", respectively.

3.2.2. Feature generation

The pre-processed tokens are further processed to generate syntactic features to describe the text. The proposed feature generation approach consists of Part-of-speech (POS) tagging and dependency parsing.

(1) POS tagging: a POS is a category of words that have similar syntactic properties. POS tagging is the process of marking up a word in a text (corpus) as corresponding to a particular POS. As an example in Fig. 3, "raise" and "forks" are tagged as VERB and NOUN, respectively.

(2) Dependency parsing: dependency syntax postulates that syntactic structure consists of relations between lexical items. In linguistics, the *head* of a phrase is the word that determines the syntactic category of that phrase and the other elements of the phrase modify the head (*head's dependents*) [26], that is, a word depends on another either if it is a complement or a modifier of the latter. The output of dependency parsing is structured as a tree, i.e., dependency tree, as illustrated in Fig. 3, where each arrow connects a head to a dependent and is typed with the name of syntactic relations (dependency labels). For example, "lower" is the head of the phrase "lower forks", which is modified by its direct object (labeled as *dobj*) "forks". "raise" is the *root* of the sentence and not depend on others.

3.2.3. Ontology-based semantic analysis

In our linguistic information extraction approach, both syntactic (POS tags, dependencies) and semantic features are generated. Semantic features are generated using an ontology-based semantic analysis method. As common regulatory rule sentences are formed with entities,

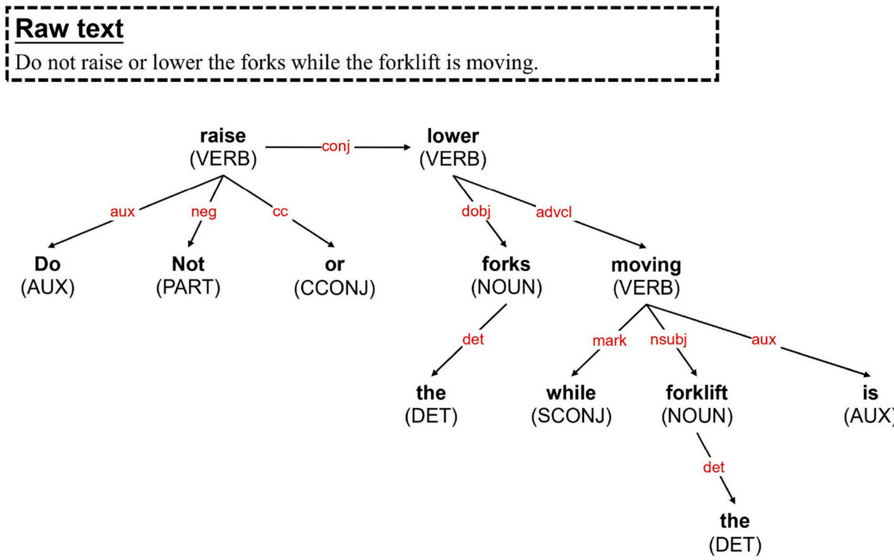


Fig. 3. An example of feature generation.

together with their attributes and the relations between each entity, the three base classes in our proposed ontology model are *Entity*, *Status*, and *Relation* (as illustrated in Fig. 4)

(1) *Entity* is the base class for modeling the various types of entities described in the regulatory rules, and its attributes are shown in Table 1. Specifically, the attribute *hasStatus* is a list that describes the status of an instance of class *Entity*, which refers to an instance of the class *Status*:

$$\forall hasStatus.Status \sqsubseteq Status \tag{1}$$

Subsequently, the attribute *hasRelation* refers to a list of instances of the class *Relation* to create relationships between instances of *Entity*.

$$\forall hasRelations.Relation \sqsubseteq Relation \tag{2}$$

The attribute *hasRequirementType* is used to specify the type of requirement: obligation (i.e., "must…"), prohibition (i.e., "must not…"), or condition (i.e., "if…then…"). Additionally, three subclasses of *Entity* are defined in ontology which inherit all attributes of the base class to further describe the entities in regulatory rule sentences: *Object* (on-site inanimate objects), *Person*, and *Area* (the place where a *Person* is working on or a *Object* is set to).

$$Object \sqsubseteq Entity \tag{3}$$

$$Person \sqsubseteq Entity \tag{4}$$

$$Area \sqsubseteq Entity \tag{5}$$

(2) *Status* is the base class used to describe the status of *Entity* instances, which is subdivided into *StaticStatus* (e.g., dismantled) and *DynamicStatus* (e.g., moving).

$$StaticStatus \sqsubseteq Status \tag{6}$$

$$DynamicStatus \sqsubseteq Status \tag{7}$$

As shown in Table 2, the attributes *hasQuantityValue* and *hasQuantityUnit* are used to specify the detailed quantity value in the a *Status* (e.g., *5 m*).

(3) *Relation* is the base class used to define the relationship between two instances of *Entity*, which is subdivided into *QuantitativeRelation* (e.g., wear) and *ComparativeRelation* (e.g., less than).

$$QuantitativeRelation \sqsubseteq Relation \tag{8}$$

$$ComparativeRelation \sqsubseteq Relation \tag{9}$$

As shown in Table 3, the attribute *hasHeads* and *hasTargets* are used to create relationships between certain instances of *Entity* to other instances of Entity.

$$\forall hasHeads.Head \sqsubseteq Entity \tag{10}$$

$$\forall hasTargets.Target \sqsubseteq Entity \tag{11}$$

Besides, for the *QuantitativeRelation* class, the attribute *hasRelationType* is used to specify the type of a *QuantitativeRelation*: modification (e.g., move), possession (e.g., wear), or loacting (e.g., on).

Additionally, to enhance the automated information extraction capability of the proposed ontology model, we adopt a gazetteer that provides the following set of term lists for the instantiation of the ontology:

(1) *PPE List*: hard hat, safety helmet, safety glasses, dust mask, full face mask, glove, welding glove, heavy-duty rubber glove, insulated glove, body harness, work shoes, boots, safety toed footwear;
(2) *Vehicle List*: car, forklift, crane, dump truck, bulldozer, front Loader, grader, trencher;
(3) *Tool List*: grinder, saw, circular saw, drill, hammer, crowbar, drill machine, polisher, vibrator, trowel;
(4) *Supply List*: ladder, power line, barrel, box, brick, block, guardrail, mid-rail, toeboard, load, fork;
(5) *Person List*: worker, construction worker, operator, observer, ordinary person;
(6) *Area List*: work area, height, heavy equipment, scaffold, construction site;
(7) *Dynamic Status List*: welding, cutting, grinding, nailing, concrete work, moving;
(8) *Static Status List*: erected, moved, dismantled, altered, height, width, volume;
(9) *Comparative Relation List*: less than, greater than, equal to, at least, at most;

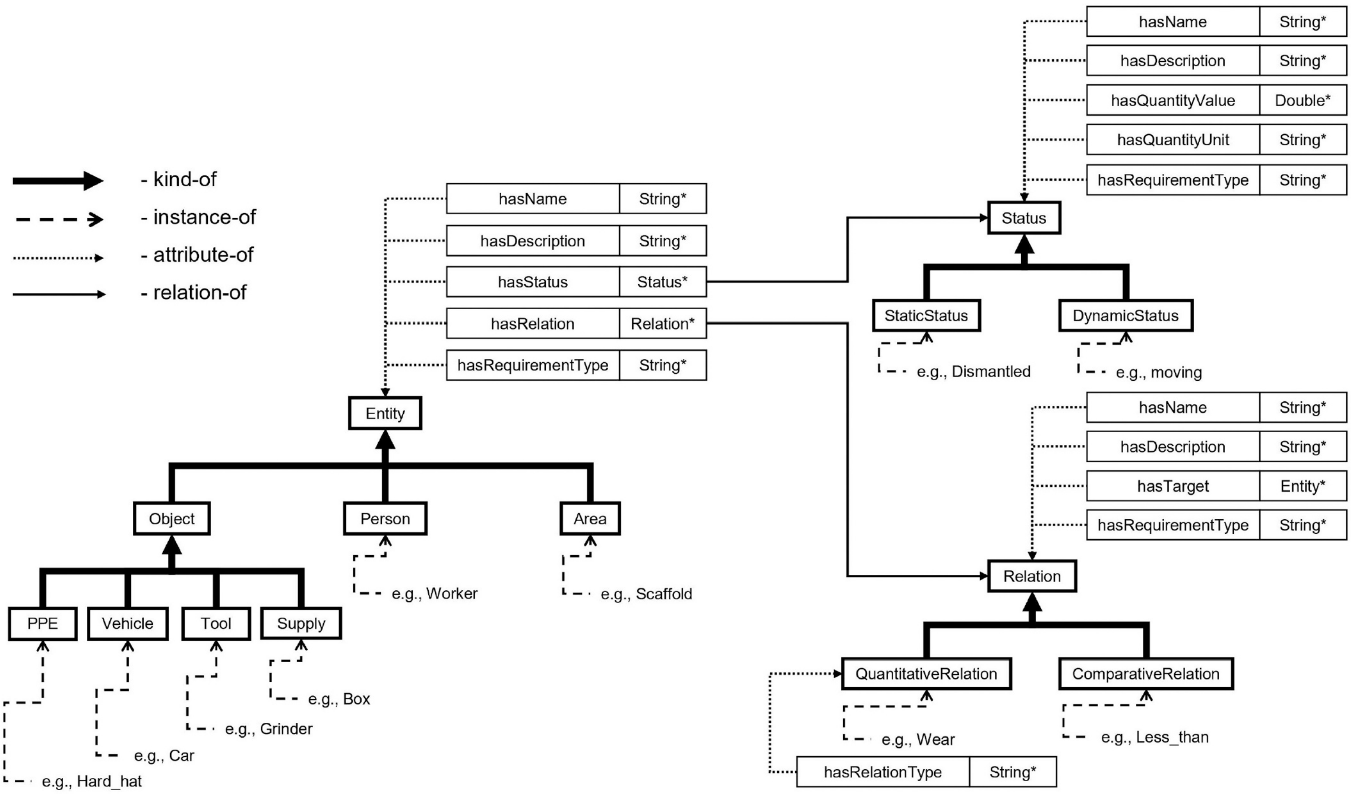


Fig. 4. The framework of the proposed ontology model.

Table 1
The attributes of the class *Entity*.

Name	Type	Description
hasName	String	The name of an entity.
hasDescription	String	The description of an entity.
hasStatus	List	The (dynamic or static) status of an entity.
hasRelations	List	The relations connected to an entity.
hasRequirementType	String	The requirement type attached to an entity in the rule sentence.

(10) *Modification Relation List*: drive, move, use, equip, provide, support, raise, lower, handle, contact;
(11) *Possession Relation List*: has, wear;
(12) *Locating Relation List*: on, from, over, enter, around;
(13) *Quantity Unit List*: foot, m, cm, kg, g, ton;

As an example, for the regulatory rule "Wear a hard hat on a construction site", the items "hard hat" and "construction site" are included in the *PPE list* and *Area list*, respectively, and the items "wear" is included in the *Possession Relations list*.

Based on the ontology modeling, we perform phrase relationship analysis on the dependency tree. The pipeline of the proposed phrasal relationship parsing is introduced in Algorithm 1. To parse relationships of each *head* with all its *dependents*, we use the Breadth-first Search (BFS) algorithm to traverse all tokens in the dependency tree *T* generated from a regulatory rule sentence.

Algorithm 1. BFS-based algorithm for phrasal relationship analysis.

At the start, the *root* token of *T* is enqueued[1] into the token queue *Q* (line 5 in Algorithm 1). *Q* contains the route along which the algorithm is currently searching. For each step, a toekn is dequeued[2] from *Q* (line 7 in Algorithm 1) while all its *dependents* are enqueued into *Q* (line 9 in Algorithm 1). Subsequently, the syntactic relation (dependency label) connecting the *dependent* to its *head* is obtained and the phrasal relationship parsing is performed as the following patterns:

(1) If the dependency label of the *dependent* is *compound* (either a noun modifying the head of noun phrase, a number modifying the head of quantifier phrase, or a hyphenated word (or a preposition) modifying the head of the prepositioanl phrase) or *amod* (adjectival modifier: an adjective or an adjective phrase that modifies the meaning of another word), then the *dependent* needs to be merged together with the *head* to form a new token (line 11 in Algorithm 1). As an example, for the regulatory rule "Use safety net systems or personal fall arrest systems.", the compound *dependent* "safety" and the adjectival modifier *dependent* "net" are merged together with the *head* "system" to form a new token "safety net systems".
(2) If the dependency label of the *dependent* is *nsubj* (nominal subject: a non-clausal constituent in the subject position of an active verb), which usually indicates an action taken by an individual in regulatory rules, then the *head* and the *dependent* are instanced as *QuantitativeRelation* and *Person*, respectively. Besides, the attribute *hasRelations* of the instance of the *dependent* is set to the instance of the *head* (line 14 in Algorithm 1). As an example, for the regulatory rule "Operators shall always wear seatbelts.", the *head* "wear" is identified as an *QuantitativeRelation* and its nominal subject "Operators" is identified as an *Person*.
(3) If the dependency label of the *dependent* is *nsubjpass* (nominal passive subject: a non-clausal constituent in the subject position

[1] Adds an item to the end of a queue.

[2] Removes and returns the item at the beginning of a queue.

```
Input:
The dependency tree, T;
The token queue, Q;
Output:
The topological set of the ontology model, S;
Q.enqueue(T.root)
while Q is not empty do
    h ⇐ Q.dequeue()
    for all d such that d ∈ h.dependents do
        Q.enqueue(d)
        if d.dep is compound ∨ d.dep is amod then
            h ⇐ merge(d, h)
        else if d.dep is nsubj then
            S.insert(QuantitativeRelation(h))
            Entity(d).hasRelations ⇐ QuantitativeRelation(h)
            S.insert(Entity(d))
        else if d.dep is nsubjpass ∨ d.dep is dobj then
            S.insert(Entity(d))
            QuantitativeRelation(h).hasTargets(Entity(d))
            S.insert(QuantitativeRelation(h))
            Entity(person).hasRelations ⇐ QuantitativeRelation(h)
            S.insert(Entity(person))
        else if d.dep is prep then
            S.insert(QuantitativeRelation(h))
            S.insert(QuantitativeRelation(d))
            QuantitativeRelation(d).hasRequirementType(Condition)
        else if d.dep is conj then
            C.insert(h, d)
        else if d.dep is neg then
            QuantitativeRelation(h).hasRequirementType(Prohibition)
            S.insert(QuantitativeRelation(h))
        else if d.dep is advcl then
            Status(d).hasRequirementType(Condition)
            S.insert(Status(d))
        else
            Skip;
        end if
    end for
end while
    return S;
```

of a passive verb), then the *head* and the *dependent* are instanced as *QuantitativeRelation* and *Entity*, respectively. Additionally, to provide more detailed description of the *dependent* token, further searching is performed in the Gazetteer lists to identify whether the *dependent* token is available in *PPE List*, *Vehicle List*, *Tool List*, *Supply List*, or *Area List* to instance *dependent* as a subclass of *Entity*. Besides, the attribute *hasRelations* of the instance of the *dependent* is set to the instance of the *head*. Subsequently, a passive object needs to be added into the phase by the instantiation of a *Person* in the ontology model (line 20 in Algorithm 1). As an example, for the regulatory rule "Safety glasses or face shields are worn when exposed to any electrical hazards including work on energized electrical systems.", the *head* "worn" is identified as an *QuantitativeRelation* and its nominal passive subject "Safety glasses" is identified as an *Entity* (further instanced as *PPE* by searching the Gazetteer lists).

(4) If the dependency label of the *dependent* is *dobj* (direct object: a noun phrase that is the accusative object of a (di)transitive verb), then the *head* and the *dependent* are instanced as *QuantitativeRelation* and *Entity*, respectively. Additionally, to provide more detailed description of the *dependent* token, further searching is performed in the Gazetteer lists to identify whether the *dependent* token is available in *PPE List*, *Vehicle List*, *Tool List*, *Supply List*, or *Area List* to instance *dependent* as a subclass of *Entity*. Subsequently, a subject needs to be added into the phase by the instantiation of a *Person* in the ontology model (line 20 in Algorithm 1). As an example, for the regulatory rule "Never enter an unprotected trench.", the *head* "enter" is identified as an *QuantitativeRelation* and its direct object "unprotected trench" is identified as an *Entity* (further instanced as *Area* by searching the Gazetteer lists).

(5) If the dependency label of the *dependent* is *prep* (prepositional modifier: any prepositional phrase that modifies the meaning of its head), then the classes of the *head* and the *dependent* in the ontology model are both identified as *QuantitativeRelation*. And *prep* in regulatory rules usually indicates a conditional clause is followed, thus *hasRequirementType* attribute of the instance of the *dependent* is set to "Condition" (line 25 in Algorithm 1). Subsequently, if the *head* had no nominal subject, a passive object needs to be added into the phase by the instantiation of a *Person* in the ontology model. As an example, for the regulatory rule "Wear a safety helmet on a construction site.", the *head* "Wear" and its prepositional modifier "on" are identified as an *QuantitativeRelation*.

(6) If the dependency label of the *dependent* is *conj* (conjunct: a dependent of the leftmost conjunct in coordination), then the *head* and its conjunct are put into a container *C* in which all instances share their attributes (line 27 in Algorithm 1). As an example, for the regulatory rule "Do not raise or lower the forks while the forklift is moving.", the *head* "raise" and its conjunct "lower" share their attributes.

(7) If the dependency label of the *dependent* is *neg* (negation modifier: an adverb that gives negative meaning to its head), then the *hasRequirementType* attribute of the instance of the *head* is set to "Prohibition" which represents a prohibition relationship in a regulatory rule (line 29 in Algorithm 1). As an example, for the regulatory rule "Never enter an unprotected trench.", the *head* "enter" has a negative modifier "Never", thus the *hasRequirementType* attribute of "enter" is set to "Prohibition".

(8) If the dependency label of the *dependent* is *advcl* (adverbial clause modifier: a clause that acts like an adverbial modifier), which in regulatory rules usually indicates a conditional clause is followed, then the *hasRequirementType* attribute of the instance of the *head* is set to "Condition" which represents a condition relationship (line 32 in Algorithm 1). As an example, for the regulatory rule "Do not raise or lower the forks while the forklift is moving.", the *head* "lower" has a adverbial clause modifier "moving", thus the *hasRequirementType* attribute of "moving" is set to "Condition".

(9) Other dependency labels are all ignored.

Table 2
The attributes of the class *Status*.

Name	Type	Description
hasName	String	The name of a status.
hasDescription	String	The description of a status.
hasQuantityValue	Double	The quantity value specified in a status.
hasQuantityUnit	String	The quantity unit specified in a status.
hasRequirementType	String	The requirement type attached to a status in the rule sentence.

Table 3
The attributes of the class *Relation*.

Name	Type	Description
hasName	String	The name of a relation.
hasDescription	String	The description of a relation.
hasHeads	List	The heads connected to a relation.
hasTargets	List	The targets connected to a relation.
hasRequirementType	String	The requirement type attached to a relation in the rule sentence.

All of the instances are stored in a topological set *S* of the ontology model, which contains the semantic information of the regulatory rules.

3.2.4. Linguistic information representation

The extracted regulatory information are first encoded in the logic representation with semantic phrases and logical connectives. The instances in the topological set *S* are transformed to semantic phrases which are defined as a triplet (e.g., (*element1*, *connection*, *element2*)). We define two types of triplets to represent the relationship between two entities or the status of an entity as shown in Table 4. For semantic phrases of entity relationships, "*element1*" or "*element2*" is a instance of *Entity* in the ontology model, which can be an instance of *Person* (e.g., worker), an instance of *Object* (e.g., glove), or an instance of *Area* (e.g., construction site), while "*connection*" is an instance of *QuantitativeRelation* which semantically connects entities with limitations. On the other hand, for semantic phrases of a entity status, "*element1*" is a instance of *Entity*, "*element2*" is a instance of *Status* (e.g., moving) in the ontology model, and "connection" in this case is set as "be".

Furthermore, we use four types of logical connectives to represent the requirement types in the regulatory rules:

(1) *If…then…* (→) is used to indicate a conditional relation that assigns the conditional requirement triplets (one or more) on the left to the instructed triplets (one or more) on its right;
(2) *Negation* (¬) is used to indicate that the requirement type of the triplet to its right is prohibition;
(3) *Logical* conjunction (∧) is used to indicate a conjunction relationship between the triplets;
(4) *Logical* disjunction (∨) is used to indicate a disjunction relationship between the triplets.

As an example, Fig. 5(b) demonstrates the logic representation of the regulatory rules in Fig. 5(a).

Table 4
Examples of phrases in the regulatory rules and the transformed triplets.

Type	Phrases	Triplets
Entity relationship	"operator wears seatbelt"	(operator, wear, seatbelt)
Entity status	"forklift is moving"	(forklift, be, moving)

Based on the logic representation (Fig. 5(b)), we use graph structures to represent regulatory information. However, even though the notion of graph representation has attracted researchers' attention for on-site scene understanding in AEC industry, approaches for representing linguistic information of regulatory rules using graph structure have not yet emerged because the pairwise relationships in traditional scene graph cannot represent the complex relationships in regulatory rules:

(1) Conditional relationships, e.g., use body harness when working on height;
(2) Prohibition relationships, e.g., never operate a grinder near face;

To this end, we have proposed an original hierarchical graph structure in our previous work to represent the linguistic information extracted from regulatory rules [24]. We further improve this hierarchical graph structure in this paper by dealing with the prohibition relationships in regulatory rules. Let $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$ be the hierarchical graph of the regulatory rules, where $\widehat{V}$ is the set of vertices to represent the elements in the semantic phrase triplets and $\widehat{E} = \{\{\mu,\nu,s,r,t\} : (\mu,\nu) \in \widehat{V}^2, \mu \neq \nu\}$ is the set of edges to represent the relationships in the semantic phrase triplets (s and r are the connections to represent a entity relationship and a entity status, respectively. t is the edge property to indicate the type of the requirements: obligation rule or prohibition rule). $\widehat{C} = \{k \rightharpoonup I : k \in \widehat{E}, I \subseteq \widehat{E}\}$ is the set of conditional relations assigning instructed triplets to conditional triplets, which are stored in a hash map. The structure of a hierarchical graph is visualized in Fig. 5(c):

(1) Dots on the inner layer represent elements of the conditional triplets;
(2) Dots on the intermediate layer represent elements of the instructed prohibition triplets;
(3) Dots on the outer layer represent elements of the instructed obligation triplets;
(4) The red lines in the circle denote the conditional relations between triplets.

In this example, the regulatory hierarchical graph is consists of the vertices $\widehat{V} = \{person, construction\ site, chemical, safety\ helmet, glove\}$, the edges $\widehat{E} = \{(person, construction\ site, on), (person, chemical, contact), (person, chemical, handle), (person, safety\ helmet, wear, obligation), (person, glove, wear, obligation)\}$, and the conditional relationships set $\widehat{C} = \{((person, construction\ site, on)) \rightharpoonup ((person, safety\ helmet, wear, obligation)), ((person, chemical, contact), (person, chemical, handle)) \rightharpoonup ((person, glove, wear, obligation))\}$.

Consequently, regulatory rules documented with natural language sentences are transformed into a calculable and structured hierarchical graph with the ability to represent complex requirements in the linguistic information of regulatory rules.

3.3. Visual information extraction

We propose a novel scene understanding approach employing scene graph as the basic notion of information representation structure to extract visual information from images, the base framework of which has been presented in our previous work [24]. As the current experimental objective is to perform automated compliance checking on regulatory rules related to the proper use of PPE and safe operation of tools in the scenes of working on height and grinder operation at construction sites, this paper mainly addresses three types of objects: head protection PPE (hard hats, safety glasses, and dust masks), grinders and body harnesses.

3.3.1. Entity extraction

An entity extractor for processing images obtained from on-site surveillance cameras is developed. For each image, individual(s) are detected, together with their body joint positions, using OpenPose [27]. Meanwhile, objects (e.g., PPE, tools) are recognized and localized by training an object detection model based on YOLOv4 [29].

(1) Individual detection: in contrast to the common vision-based on-site occupational hazards identification approaches [11–15], we employ an individual detection model to specify individual features. This strategy makes our entity extractor be more robust with viewpoint changes and different individual postures. The workers' postures are characterized by extracting the body joint positions of the individuals in the image using OpenPose [27], which is the state-of-the-art model for estimating human pose in images. Instead of the conventional top-down approaches which employ a person detector and performing single-person pose estimation for each detected person, OpenPose provides a bottom-up approach by detecting all body parts in the image and associating them with different person. OpenPose provides the positions of 18 body joints (pre-trained using MSCOCO 2016 keypoints challenge dataset [28], see Fig. 6). The choice of OpenPose is motivated for its functionality on RGB images or videos taken by on-site surveillance cameras. This provides a huge benefit in comparison to the skeletal tracking capability of RGB-D devices which depend on depth information. Besides, in contrast to the top-down approaches, the inference time of OpenPose is invariant to the number of people in the image and able to provide real-time performance.

(a) (b) (c)

Fig. 5. An example of the regulatory rules and its hierarchical graph. (a) The regulatory rules are decomposed and transformed to (b) the logic representation; (c) Hierarchical graph.

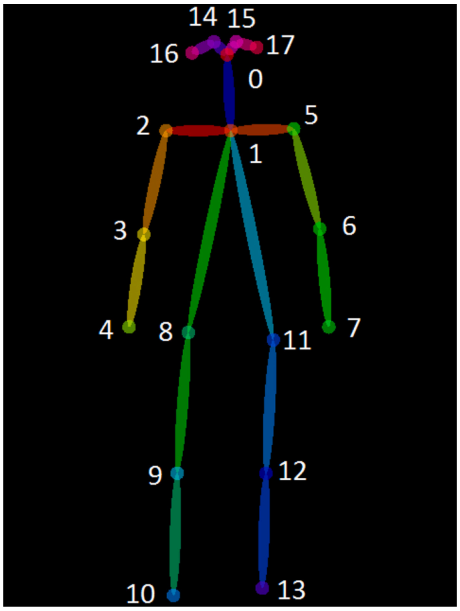


Fig. 6. Output format of OpenPose.

(2) Object detection: objects recognition and localization are carried out using a light-weight object detection model. To achieve the optimal trade-off between speed and accuracy on a multi-class object detection task, we suggest that one-stage approaches are preferable options, e.g., YOLO and its variants [30–32], which skips the region proposal stage and runs detection directly over a dense sampling of possible locations, and it leads to a simpler architecture and makes it extremely fast. In contrast to the one-stage object detection model deployed in our previous works, i.e., YOLOv3 [32], this paper adopts YOLOv4 [29], a newly improved model of the YOLO family, for the PPE detection task. YOLOv4 uses a deep network architecture, i.e., CSPDarknet53 [33], as the backbone for feature extraction. Meanwhile, spatial pyramid pooling (SPP) block [34] is added to CSPDarknet53 to increase the receptive field and to separate the most significant contextual features. Furthermore, in contrast to the Feature Pyramid Network (FPN) [35] used in YOLOv3 for object detection, YOLOv4 uses PANet [36] as a parameter aggregation method for different detector levels. These new features of YOLOv4 improve the efficiency and performance for multi-class object detection, which make YOLOv4 a reasonable option for real-time processing for industrial purpose. Compared with models such as Faster R-CNN, SSD, etc., which are commonly used to perform visual tasks in the AEC domain [11–15], YOLOv4 is one of the most performing novel object detection models currently available. Additionally, YOLOv4 provides model scaling functionality, which makes it adaptable to varying on-site hardware devices with different compute capabilities (general GPUs, low-end GPUs, or high-end GPUs) and thus increases the usability of the occupational hazards identification system.

3.3.2. *Individual-object relationship analysis*

We first associate the detected objects with the detected individuals, with the aim of providing a prior knowledge for individual-object relationship analysis and reducing computational complexity. A weighted bipartite graph is constructed to represent the detected entities and we perform individual-object association as a minimum weighted matching in bipartite graphs. Subsequently, we analyze the individual-object relationship on the associated individual-object pairs M, which address three types of objects in this paper:

(1) Head protection PPE (hard hats, safety glasses, and dust masks).

For each associated individual and head protection PPE $\{i^*, j^*\} \in M$ their relationship is identified by measuring a distance. Considering the fact that both the individual and object detection model acquire position information based on image pixel coordinates that vary with the distance relative to the camera, we take advantage of the Euclidean distance among detected neck keypoint (body parts 1 in Fig. 6) and hip keypoints (body parts 8 and 11 in Fig. 6) of i^* as a dynamic reference threshold, which will keep changing synchronously when the distance between the individual and the camera changes:

$$\beta_{i^*\leftrightarrow j^*} = max\left(\sqrt{\left(x_{i^*}^{(1)} - x_{i^*}^{(8)}\right)^2 + \left(y_{i^*}^{(1)} - y_{i^*}^{(8)}\right)^2},\ \sqrt{\left(x_{i^*}^{(1)} - x_{i^*}^{(11)}\right)^2 + \left(y_{i^*}^{(1)} - y_{i^*}^{(11)}\right)^2}\right)\cdot\gamma \tag{12}$$

where γ is the scaling coefficient to strike the relationship analysis for different head protection PPE, which was obtained based on ergonomic analysis and statistical experiments on randomly selected samples from the validation dataset. For hard hats, safety glasses, dust masks, and full-face masks, γ is set to 0.8, 0.7, 0.6, 0.6, respectively.

If the Euclidean distance between the position $(x_j{}^*, y_j{}^*)$ of the bounding box of j^* and detected neck keypoint (body parts 1 in Fig. 6) of i^* is smaller than the reference threshold $\beta_{i^*\leftrightarrow j^*}$, then the relationship between the i^* and j^* is created; otherwise, even though j^* is associated with i^*, no relationship is created between them:

$$c_{i^*\leftrightarrow j^*} = \begin{cases} \text{“}wear\text{”} & , if\ d_h(i^*, j^*) < \beta_{i^*\leftrightarrow j^*} \\ N/A, & otherwise \end{cases} \tag{13}$$

where

$$d_h(i^*, j^*) = \sqrt{\left(x_{i^*}^{(1)} - x_{j^*}\right)^2 + \left(y_{i^*}^{(1)} - y_{j^*}\right)^2} \tag{14}$$

and $c_i{}^*{}_{\leftrightarrow j}{}^*$ indicates the connection to create the relationship between i^* and j^* as a semantic phrase $(i^*, c_i{}^*{}_{\leftrightarrow j}{}^*, j^*)$ (e.g., (*person, wear, hard hat*) or not.

(2) Grinders

Currently in this work, two regulatory rules related to grinder proper use are addressed to create a individual-grinder relationship:

(A) “Always use two hands when operating a grinder”.

Let $B_g{}^* = (x_g{}^*, y_g{}^*, w_g{}^*, h_g{}^*)$ be the detected bounding box of a grinder g^* which is associated with the detected individual i^*. Firstly, the Euclidean distance from the left wrist keypoint and right wrist keypoint (body parts 7 and 4 in Fig. 6) of i^* to the position $(x_g{}^*, y_g{}^*)$ of g^* is calculated (Fig. 7):

$$\begin{aligned} d_l(i^*, g^*) &= \sqrt{\left(x_{i^*}^{(7)} - x_{g^*}\right)^2 + \left(y_{i^*}^{(7)} - y_{g^*}\right)^2} \\ d_r(i^*, g^*) &= \sqrt{\left(x_{i^*}^{(4)} - x_{g^*}\right)^2 + \left(y_{i^*}^{(4)} - y_{g^*}\right)^2} \end{aligned} \tag{15}$$

If the grinder is close enough to the wrists, then the individual is identified as holding the grinder:

Fig. 7. Relationship identification strategies to address the rule "Always use two hands when operating a grinder".

$$h_{i^*\leftrightarrow g^*} = \begin{cases} 1 & , if\ d_l(i^*,g^*) < \beta_{i^*\leftrightarrow g^*}\ or\ d_r(i^*,g^*) < \beta_{i^*\leftrightarrow g^*} \\ 0 & , otherwise \end{cases} \quad (16)$$

where $\beta_{i^*\leftrightarrow g^*}$ is the reference threshold calculated based on the size of the bounding box of g^*:

$$\beta_{i^*\leftrightarrow g^*} = max(w_{g^*}, h_{g^*}) \quad (17)$$

If $h_{i^*\leftrightarrow g^*} = 1$, then relationship identification needs to be further performed to identify whether the individual i^* is holding the grinder g^* using single hand or two hands. It's known that when an object is holding by two hands the distance between the wrists is small. Thus, the relationship between i^* and g^* is identified as follows:

$$c_{i^*\leftrightarrow h_{i^*}^*}, h_{i^*}^*, c_{h_{i^*}^*\leftrightarrow g^*} = \begin{cases} \text{"use", "two hands", "operate"} & , if\ d_{l\leftrightarrow r}(i^*) < \beta_{i^*\leftrightarrow g^*} \\ \text{"use", "single hand", "operate"} & , otherwise \end{cases} \quad (18)$$

where h_i^{**} indicates the hands status (e.g., two hands, single hand) when operating a grinder while $c_{i^*\leftrightarrow hi^{**}}$ and $c_{hi^{**}\leftrightarrow g^*}$ create the relationship between i^* and h_i^{**}, h_i^{**} and g^*, as semantic phrases (i^*, $c_{i^*\leftrightarrow hi^{**}}, h_i^{**}$), ($h_i^{**}, c_{hi^{**}\leftrightarrow g^*}, g^*$), respectively (e.g., the relationship (*person, use, two hands*), (*two hands, operate, grinder*) is created in Fig. 7).

(B) "Never operate a grinder near face".

For the detected individual i^* and the associated grinder g^*, the Euclidean distance from the neck keypoint (body part 1 in Fig. 6) of i^* to the position (x_{g^*}, y_{g^*}) of the bounding box of g^* is calculated:

$$d_n(i^*,g^*) = \sqrt{\left(x_{i^*}^{(1)} - x_{g^*}\right)^2 + \left(y_{i^*}^{(1)} - y_{g^*}\right)^2} \quad (19)$$

If the grinder is close enough to the neck, then the relationship between the grinder and the face of individual is created:

$$c_{i^*\leftrightarrow g^*}^{face} = \begin{cases} \text{"near"} & , if\ d_n(i^*,g^*) < \beta_{i^*\leftrightarrow g^*} \\ N/A & , otherwise \end{cases} \quad (20)$$

where $\beta_{i^*\leftrightarrow g^*}$ is the reference threshold calculated based on the Euclidean distance among detected neck keypoint (body parts 1 in Fig. 6) and hip keypoints (body parts 8 and 11 in Fig. 6) of i^* with $\gamma = 0.3$:

$$\beta_{i^*\leftrightarrow g^*}^{face} = max\left(\sqrt{\left(x_{i^*}^{(1)} - x_{i^*}^{(8)}\right)^2 + \left(y_{i^*}^{(1)} - y_{i^*}^{(8)}\right)^2}, \sqrt{\left(x_{i^*}^{(1)} - x_{i^*}^{(11)}\right)^2 + \left(y_{i^*}^{(1)} - y_{i^*}^{(11)}\right)^2}\right)\cdot\gamma \quad (21)$$

and $c_{i^*\leftrightarrow g^*}^{face}$ indicates the connection to create the relationship between the face of i^* and g^* as a semantic phrase ($face, c_{i^*\leftrightarrow g^*}^{face}, g^*$) (e.g., the relationship (*face, near, grinder*) is created in Fig. 8)

(3) Body harnesses:

Let $B_{h^*} = (x_{h^*}, y_{h^*}, w_{h^*}, h_{h^*})$ be the bounding box of a detected body harnesses h^* which is associated with the detected individual i^*. Their relationship is identified by performing the geometric relationship analysis for the key point $(x_{i^*\leftrightarrow h^*}^k, y_{i^*\leftrightarrow h^*}^k)$, the geometric center of the neck keypoint (body part 1 in Fig. 6) and hip keypoints (body parts 8 and 11 in Fig. 6) of i^*, which is given as follows:

$$x_{i^*\leftrightarrow h^*}^k = \frac{x_{i^*}^{(1)} + x_{i^*}^{(8)} + x_{i^*}^{(11)}}{3}$$
$$y_{i^*\leftrightarrow h^*}^k = \frac{y_{i^*}^{(1)} + y_{i^*}^{(8)} + y_{i^*}^{(11)}}{3} \quad (22)$$

If $(x_{i^*\leftrightarrow h^*}^k, y_{i^*\leftrightarrow h^*}^k)$ is in the bounding box B_{h^*}, then the relationship between i^* and h^* is created; otherwise, even though h^* is associated with i^*, no relationship is created between them:

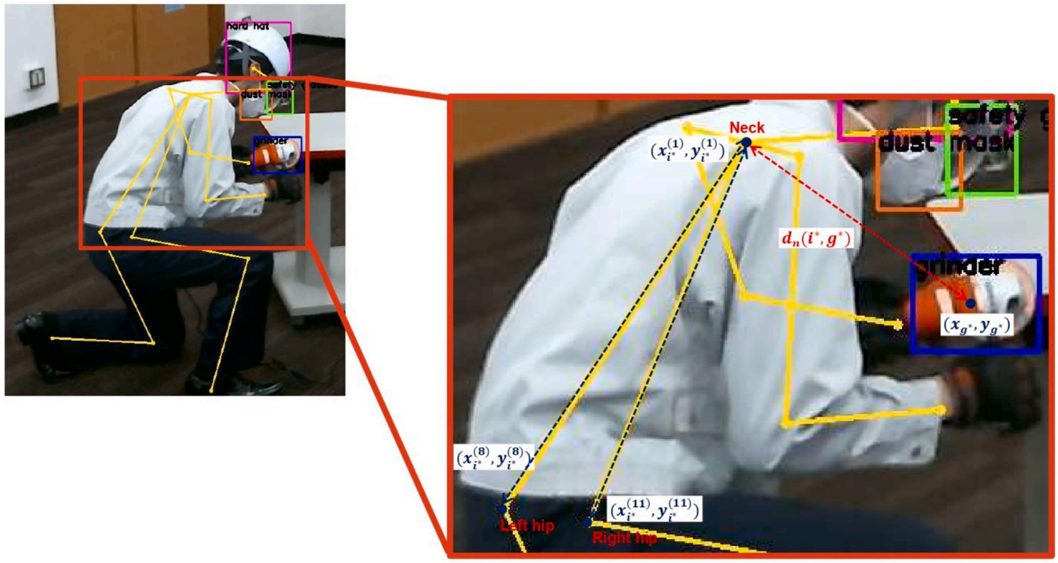


Fig. 8. Relationship identification strategies to address the rule "Never operate a grinder near face".

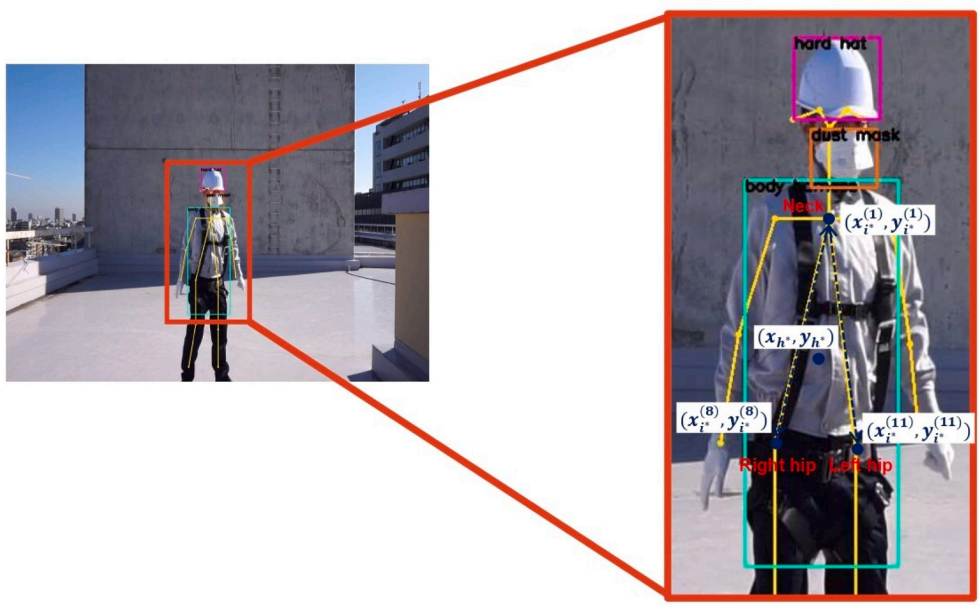
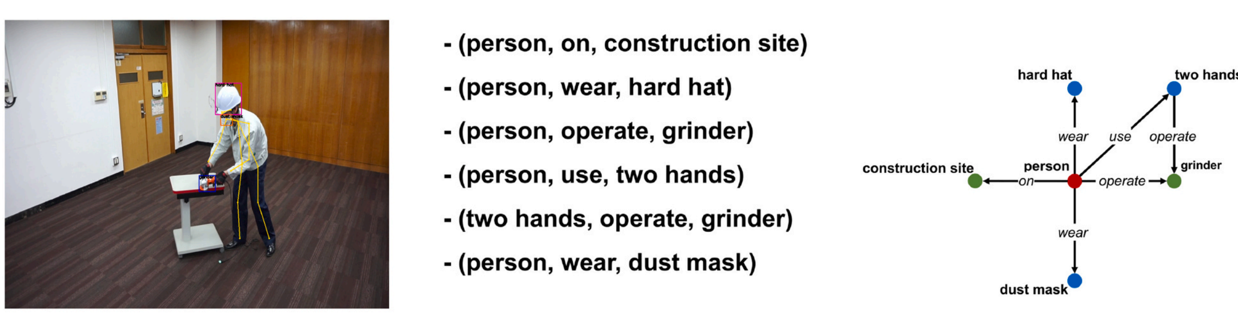


Fig. 9. The individual-body harnesses relationship identification strategies.



(a) (b) (c)

Fig. 10. An example of on-site image and its scene graph. (a) On-site image; (b) Individual-object relationships extracted from (a); (c) Scene graph $G(V,E)$.

$$c_{i^*\leftrightarrow h^*} = \begin{cases} \text{“wear”} \quad , if & x^k_{i^*\leftrightarrow h^*} \in \left[x_{h^*} - \dfrac{w_{h^*}}{2}, x_{h^*} + \dfrac{w_{h^*}}{2}\right] \; and \\ & y^k_{i^*\leftrightarrow h^*} \in \left[y_{h^*} - \dfrac{h_{h^*}}{2}, y_{h^*} + \dfrac{h_{h^*}}{2}\right] \\ N/A \quad , otherwise & \end{cases} \tag{23}$$

where $c_{i^*\leftrightarrow h^*}$ indicates the connection to create the relationship between i^* and h^* as a semantic phrase $(i^*, c_{i^*\leftrightarrow h^*}, h^*)$ (e.g., the relationship (*person*, *use*, *body harness*)) is created in Fig. 9).

3.3.3. Visual information representation

Based on the semantic phrases established by individual-object relationship analysis, we generate a scene graph for image information representation for each image. Given a scene graph $G(V,E)$ of an on-site image, where V is the set of vertices to represent the objects in the semantic phrase triplets and $E = \{\{\mu,\nu,\tau\} : (\mu,\nu) \in V^2, \mu \neq \nu\}$ is the set of edges to represent the individual-object relationships in the image (τ is the connections to represent a entity relationship or a entity status). In the example in Fig. 10, the scene graph $G(V,E)$ (Fig. 10(c)) of the on-site image (Fig. 10(a)) is consists of the vertices $V = \{person, hard\ hat, dust\ mask, grinder\}$, the edges $E = \{(person, hard\ hat, wear), (person, dust\ mask, wear), (person, grinder, operate), (person, two\ \ hands, use), (two\ \ hands, grinder, operate)\}$. As this example shows, the proposed visual information extraction module enables the generation of graph structures containing scene information (individuals, objects, etc.) through automated perception and achieves scene understanding. Notably, on the basis of the proposed approach, the task of visual information extraction is also extendable to other scenes by increasing the categories of object detection together with geometric feature analysis. Further discussion on extendability of the proposed approach in different scenes is given in Appendix B.

3.4. Automated reasoning for hazards identification

3.4.1. Relevant regulatory rules extraction

To specify the reasoning targets and reduce the computational complexity, the relevant regulatory rules for the scene information of the on-site image need to be first extracted.

As processed in Algorithm 2, given a scene graph $G(V,E)$ (Fig. 11(a)) of on-site image, where V is the set of vertices to represent the detected entities in the obtained on-site images and $E = \{\{\mu,\nu,\tau\} : (\mu,\nu) \in V^2, \mu \neq \nu\}$ is the set of edges to represent the individual-object relationships in the image, the conditional relationship triplets are extracted by searching a subset $V' \subseteq V$ which is the keys of the conditional relation set $\widehat{C}$ in $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$ (Fig. 11(b)). As the example showed in Fig. 11, $V' = \{(person, grinder, operate), (person, construction\ \ site, on)\}$. Subsequently, the corresponding vertices $\widehat{V}'$ and edges $\widehat{E}'$ are retrieved by traversing $\widehat{C}$ and a subgraph $\widehat{G}'(\widehat{V}',\widehat{E}')$ (Fig. 11(c)) is created from $\widehat{G}$. $\widehat{G}'$ contains all regulatory rules for the situation of the on-site image and is further used for hazards identification.

Algorithm 2. Relevant regulatory rules extraction algorithm.

3.4.2. Hazards identification

Based on the extracted relevant regulatory rules graph $\widehat{G}'(\widehat{V}',\widehat{E}')$, reasoning for hazards identification is performed by checking compliance of prohibition and obligation regulatory rules. Pruning is first performed on $\widehat{G}'(\widehat{V}',\widehat{E}')$ to extract the prohibition regulatory rules subgraph $\widehat{G}'_P(\widehat{V}'_P,\widehat{E}'_P)$ and the obligation regulatory rules subgraph $\widehat{G}'_O(\widehat{V}'_O,\widehat{E}'_O)$. The processing of pruning is described in Algorithm 3. As demonstrated in Fig. 12, given a relevant regulatory rules graph $\widehat{G}'(\widehat{V}',\widehat{E}')$, where $\widehat{V}' = \{person, construction\ site,$

$$hard\ hat, dust\ mask, safety\ glasses, grinder, face, two\ hands\}$$

and $\widehat{E}' = \{(person, construction\ site, on), (person, hard\ \ hat, wear, obligation), (person, dust\ \ mask, wear, obligation), (person, grinder, operate), (person, safety\ \ glasses, wear, obligation), (person, two\ \ hands, use), (two\ \ hands, grinder, operate, obligation), (grinder, face, near, prohibition)\}$, by performing pruning the prohibition regulatory rules subgraph $\widehat{G}'_P(\widehat{V}'_P,\widehat{E}'_P)$ is extracted where $\widehat{V}'_P = \{person, grinder, face\}$ and $\widehat{E}'_P = \{(grinder, face, near, prohibition)\}$, together with the obligation regulatory rules subgraph $\widehat{G}'_O(\widehat{V}'_O,\widehat{E}'_O)$ where $\widehat{V}'_O = \{person, hard\ hat, dust\ mask, safety\ \ glasses, grinder, two\ \ hands\}$ and $\widehat{E}'_O = \{(person, hard\ hat, wear, obligation), (person, dust\ \ mask, wear, obligation), (person, safety\ \ glasses, wear, obligation), (person, two\ \ hands, use), (two\ hands, grinder, operate, obligation)\}$.

Algorithm 3. Relevant regulatory rules graph pruning algorithm.

Prohibition regulatory rules reasoning is performed based on compliance checking between the scene graph $G(V,E)$ of on-site image and the prohibition regulatory rules subgraph $\widehat{G}_P{}'(\widehat{V}_P{}',\widehat{E}_P{}')$. As processed in Algorithm 4, if an edge $\widehat{e}_P{}'$ of $\widehat{E}_P{}'$ exists in E, which means a prohibition entities relationship exists in the on-site image scene, then

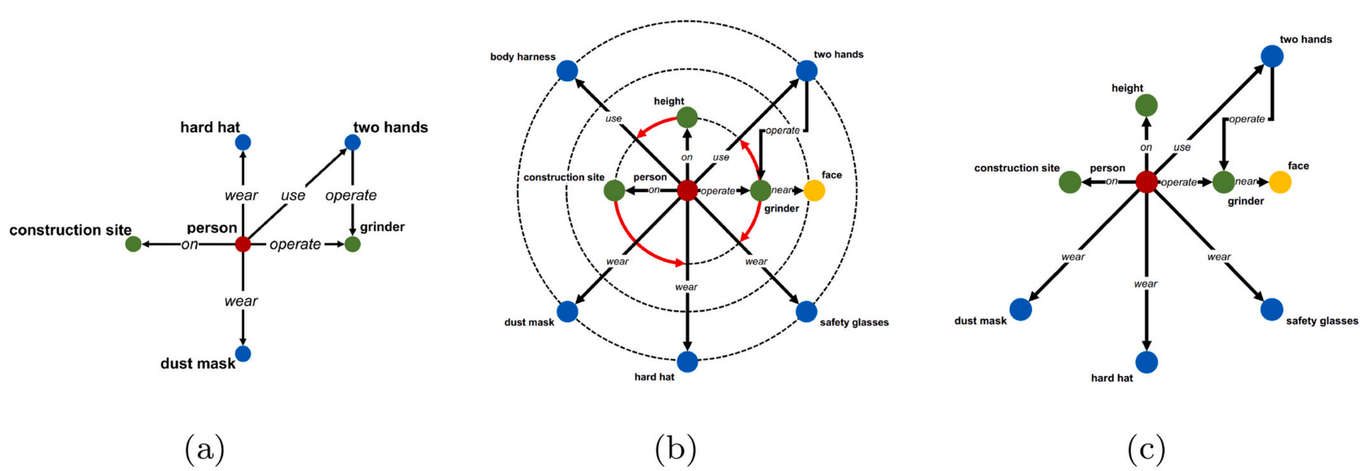


Fig. 11. (c) is the relevant rules graph $\widehat{G}'(\widehat{V}',\widehat{E}')$ extracted from (b) regulatory rules hierarchical graph $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$ and contains all regulatory rules for the situation of (a) on-site image scene graph $G(V,E)$.

Input:

On-site image's scene graph, $G(V, E)$;

Regulatory rules hierarchical graph, $\hat{G}(\hat{V}, \hat{E}, \hat{C})$;

Output:

Relevant rules graph $\hat{G}'\left(\hat{V}', \hat{E}'\right)$;

for all e such that $e \in E$ **do**

    if $\hat{C}.containsKey(e)$ **then**

        $\hat{E}'.insert(e)$

        $\hat{V}'.insert(e.head, e.target)$

        $ins \leftarrow \hat{C}.get(e)$

        for all i such that $i \in ins$ **do**

            $\hat{E}'.insert(i)$

            $\hat{V}'.insert(i.head, i.target)$

        end for

    end if

end for

return $\hat{G}'\left(\hat{V}', \hat{E}'\right)$

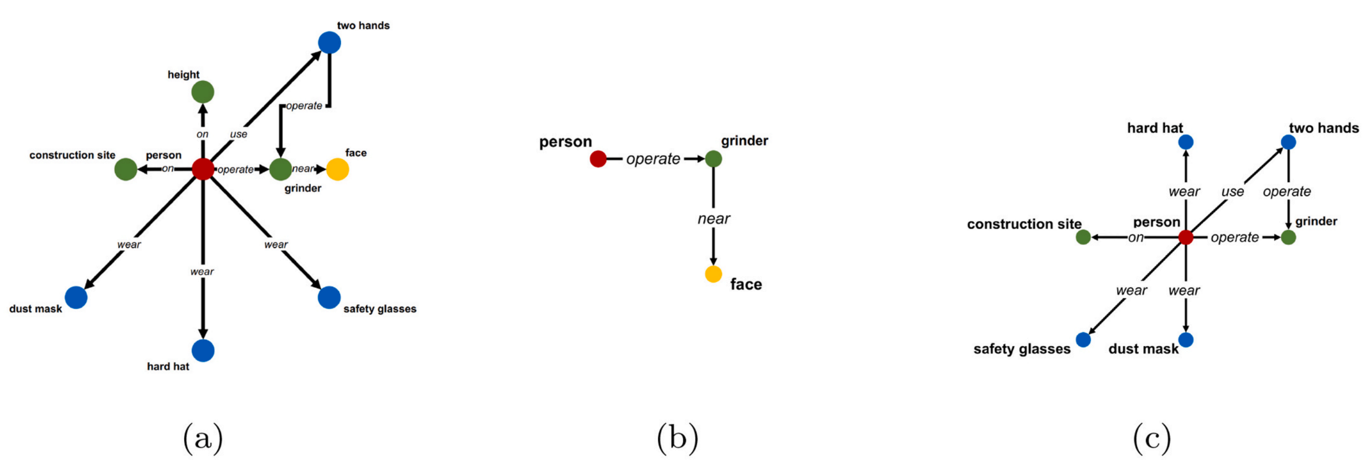


Fig. 12. By performing pruning on (a) the relevant regulatory rules graph $\widehat{G}'(\widehat{V}', \widehat{E}')$, (b) the prohibition regulatory rules subgraph $\widehat{G}_P'(\widehat{V}_P', \widehat{E}_P')$ and (c) the obligation regulatory rules subgraph $\widehat{G}_O'(\widehat{V}_O', \widehat{E}_O')$ are extracted.

$\widehat{e}_P'$ is extracted as a violated regulatory prohibition rule and the on-site image scene is hence identified as hazardous.

Algorithm 4. Prohibition regulatory rules reasoning algorithm.

The obligation regulatory rules reasoning for hazards identification of the on-site image is performed based on the isomorphism between $G(V,E)$ and $\widehat{G}_O'(\widehat{V}_O', \widehat{E}_O')$. In graph theory, an isomorphism is a mapping between two graph structures of the same type that can be reversed by an inverse mapping. $G(V,E)$ is isomorphic to $\widehat{G}_O'(\widehat{V}_O', \widehat{E}_O')$, if there exists a bijective function $f: V \rightarrow \widehat{V}_O'$ such that $\forall u, \ v \in V$, $(u,v) \in E \leftrightarrow (f(u), f(v)) \in \widehat{E}_O'$, which is denoted as $G \cong \widehat{G}_O'$ [37]. Otherwise, $G(V,E)$ is non-isomorphic to $\widehat{G}_O'(\widehat{V}_O', \widehat{E}_O')$ and the violated obligation regulatory rules $H_O = \{(\mu, \nu, \tau) \in \widehat{E}_O', (\mu, \nu, s, r) \notin E\}$ from the on-site image are identified (Algorithm 5). In the case of $G(V,E)$ in Fig. 11(a) and $\widehat{G}_O'(\widehat{V}_O', \widehat{E}_O')$ in Fig. 12(c), $H_O = (person, safety\ glasses, use)$ which represents the hazard that the person is not wearing safety glasses when operating the grinder.

Input:

Relevant regulatory rules graph, $\hat{G}'\left(\hat{V}', \hat{E}'\right)$;

Output:

Prohibition regulatory rules subgraph, $\hat{G_P}'\left(\hat{V_P}', \hat{E_P}'\right)$;

Obligation regulatory rules subgraph, $\hat{G_O}'\left(\hat{V_O}', \hat{E_O}'\right)$;

for all $\hat{e}'$ such that $\hat{e}' \in \hat{E}'$ **do**
    if $\hat{e}'.requirementType$ is $prohibition$ **then**
        $\hat{E_P}'.insert(\hat{e}')$
        $\hat{V_P}'.insert(\hat{e}'.head, \hat{e}'.target)$
    else if $\hat{e}'.requirementType$ is $obligation$ **then**
        $\hat{E_O}'.insert(\hat{e}')$
        $\hat{V_O}'.insert(\hat{e}'.head, \hat{e}'.target)$
    end if
end for
    return $\hat{G_P}'\left(\hat{V_P}', \hat{E_P}'\right), \hat{G_O}'\left(\hat{V_O}', \hat{E_O}'\right)$

Input:

On-site image's scene graph, $G(V, E)$;

Prohibition regulatory rules subgraph $\hat{G_P}'\left(\hat{V_P}', \hat{E_P}'\right)$;

Output:

Violated prohibition regulatory rules; H_P

for all $\hat{e_P}'$ such that $\hat{e_P}' \in \hat{E_P}'$ **do**
    if $\hat{e_P}'$ in E **then**
        $H_P.insert(\hat{e_P}')$
    end if
end for
    return H_P

Algorithm 5. Obligation regulatory rules reasoning algorithm.

4. Experiments and results

The purpose of the current experiments is to extract the key information of the target regulatory rules using the proposed linguistic information extraction module, followed by scene understanding of the images in the testing dataset using the proposed visual information extraction module, which finally performs a compliance checking of the images based on the extracted key information of the regulatory rules to identify potential occupational hazards.

4.1. Experimental description

4.1.1. Target regulatory rules

To demonstrate the validity of the proposed linguistic information extraction approach in this paper, we selected six regulatory rules from

Input:

On-site image's scene graph, $G\,(V, E)$;

Obligation regulatory rules subgraph $\hat{G_O}'(\hat{V_O}', \hat{E_O}')$;

Output:

Violated obligation regulatory rules; H_O

for all $\hat{e_O}'$ such that $\hat{e_O}' \in \hat{E_O}'$ **do**

  if $\hat{e_O}'$ not in E **then**

    $H_O.insert(\hat{e_O}')$

  end if

end for

  return H_O

Table 5
Information of collected training dataset.

	Number of real-world image samples	Number of web-mined image samples	Total
Hard hat	3003	1322	4325
Dust mask	3099	186	3285
Safety glasses	2180	115	2305
Body harness	347	135	482
Grinder	2005	355	2360
Overall	10,634	2113	12,747

on-site occupational safety guides in Japan to perform the experiments:

(1) “Wear a hard hat on a construction site.”
(2) “Wear a dust mask on a construction site.”[3]
(3) “Use body harness when working on height.”
(4) “Wear a safety glasses when operating a grinder.”
(5) “Always use two hands when operating a grinder.”
(6) “Never operate a grinder near face.”

4.1.2. Image datasets

To create a training dataset for object detection, we collected images of hard hats, dust masks, safety glasses, body harness, and grinders from two sources: real-world images and web-mined images. Real-world images were obtained using a SONY α5000 digital camera at six different places (including construction sites, school campus, and experimental rooms) under different environmental conditions (e.g., level of illumination, visual range) and the obtained images are all 1440 × 1080 in resolution. Web-mined images were retrieved by search engines using a web crawler. The resolution of these collected web images ranges from 150 × 150 to 4896 × 3672. A total of 6029 images containing 12,747 objects were collected and annotated as listed in Table 5.

Furthermore, to address compliance with these six regulatory rules, we created a testing dataset that contains two scenes: working on height and operating a grinder. Seven volunteers were instructed to perform different behaviors wearing different PPEs. The testing images were taken in different contexts, and the individuals in the images were at different distances from the camera and in different postures, which made the identification of the model more difficult and more meaningful for verifying its validity and robustness. For the scene of “working on height”, we captured at three different distances (3 m, 5 m, 7 m) from the camera on the rooftop of an eight-floor building, and randomly selected 1500 images from the collected images sequence at each distance as the test samples. Also, for the scene of “operating a grinder”, we

Table 6
Information of the collected testing dataset of the scene “working on height”.

Distance (m)	Number of images	Related PPE	Number of positive samples	Number of negative samples
3	1500	Hard hat	866	634
		Dust mask	818	682
		Safety belt	635	865
5	1500	Hard hat	900	600
		Dust mask	750	750
		Safety belt	750	750
7	1500	Hard hat	900	600
		Dust mask	775	725
		Safety belt	725	775

Table 7
Information of the collected testing dataset of the scene “operating a grinder”.

Number of images	Related PPE / tool	Number of positive samples	Number of negative samples
1500	Hard hat	735	765
	Safety glasses	643	857
	Dust mask	1072	428
	Grinder	443	783[1] 467[2]

[1] operate a grinder using single hand.
[2] operate a grinder near face.

[3] At some construction sites in the Fukushima area of Japan that dispose of radioactive waste, workers are required to wear dust masks to prevent inhalation of radioactive dust.

Table 8
Defination of TP, FP, and FN.

	Predicted	Ground truth
TP	Proper use (safe)	Proper use (safe)
FP	Proper use (safe)	Improper use (hazardous)
FN	Improper use (hazardous)	Proper use (safe)

also demonstrated the operation indoors and near the pipeline, and selected a random sample of 1500 images for the test. The collected testing dataset of grinder operating includes normal operation and two types of hazardous operation (operating a grinder using single hand and operating a grinder near face). The details are provided in Table 6 and Table 7, where positive samples refer to individuals who follow the regulatory rules, and negative samples refer to individuals who do not follow regulatory rules.

4.1.3. Evaluation metrics

Precision and recall were adopted to evaluate the performance of the proposed approach:

$$Precision = \frac{TP}{TP + FP} \tag{24}$$

$$Recall = \frac{TP}{TP + FN} \tag{25}$$

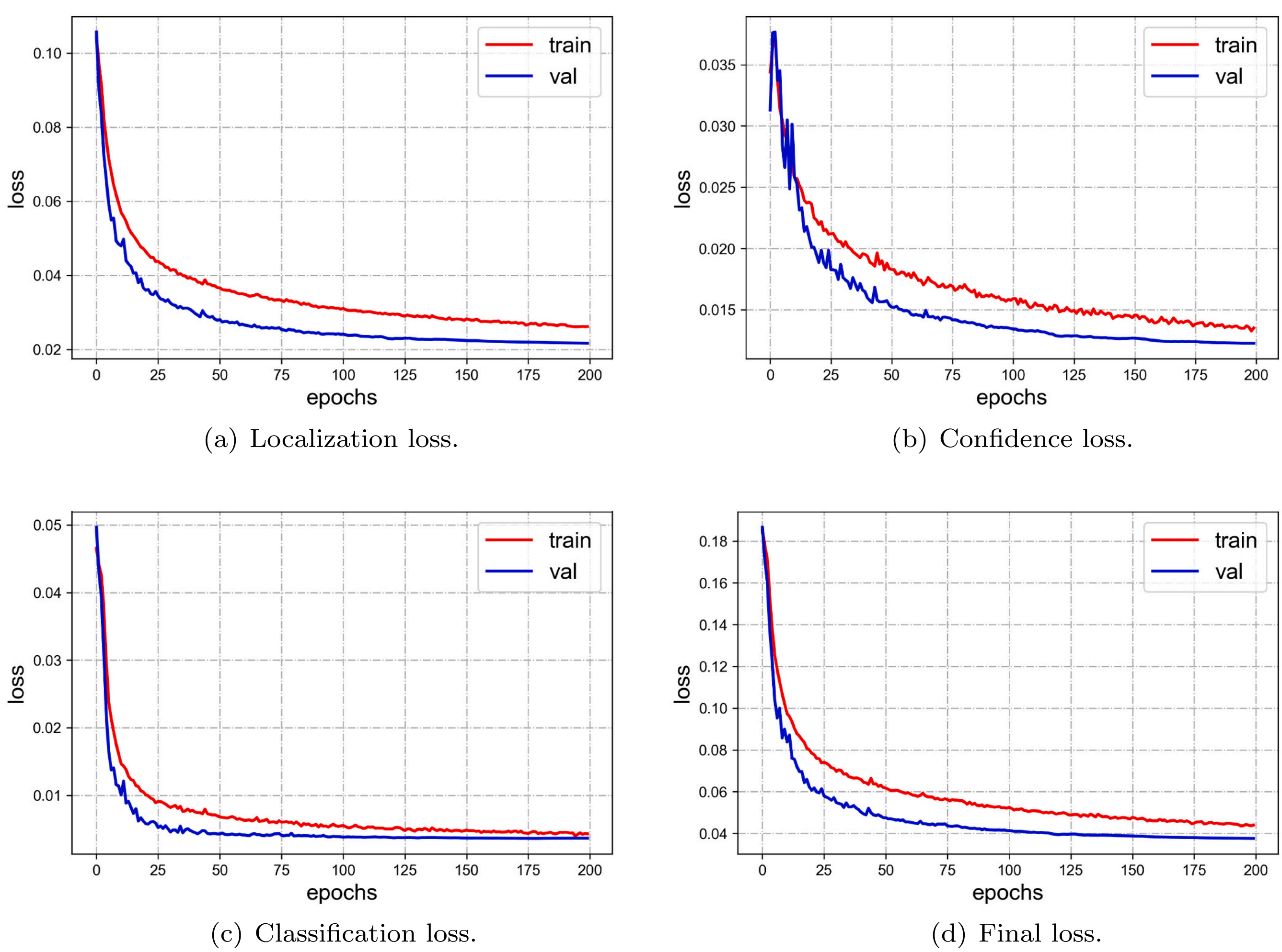
where *TP* (true positive) is defined as the number of correct identification of individuals who are properly using PPE/grinder, *FP* (false positive) is the number of wrong identification of individuals who are properly using PPE/grinder, while *FN* (false negative) is the number of the ground truth not identified as defined in Table 8.

4.2. Regulatory rules processing system development

Based the linguistic information extraction approach introduced above, we have developed a novel automated regulatory rules processing system. The open-source libraries *spaCy* [38], *networkx* [39], *graphviz* [40] are deployed for feature generation, graph structure generation, visualization, respectively.

4.3. Object detection model training

We build the YOLOv4-CSP (CSPDarknet53 as the backbone for feature extraction) model using PyTorch and initialize the model based on the pre-trained weights on the MSCOCO 2017 object detection dataset [28]. We train the YOLOv4 model for 200 epochs by stochastic gradient descent (SGD), and throughout training we use a batch size of 8, a momentum of 0.9 and a decay of 0.0005. The learning rate is initialized to $1e-2$ and is decreasing following the cosine function. 80% samples in the training dataset are used for training the YOLOv4 model, and the remaining 20% samples are reserved for validating the training process to prevent overfitting. The training and validating losses are plotted in Fig. 13. Some detected examples from the validating dataset are visualized in Fig. 14.



(a) Localization loss. (b) Confidence loss. (c) Classification loss. (d) Final loss.

Fig. 13. Training and validating losses.

Fig. 14. Object detection examples on the validating dataset.

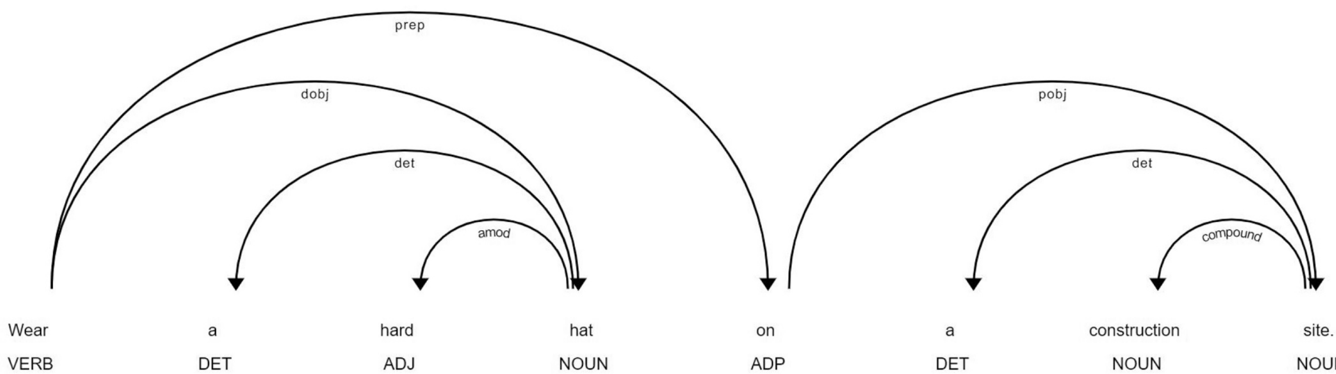


Fig. 15. The dependency tree of regulatory rule: "Wear a hard hat on a construction site."

All experiments are performed on a machine with Intel Core i7-7820× (8 cores, 3.6GHz), 32GB DDR4 SDRAM RAM, NVIDIA GeForce GTX 1080 Ti GPU (11GB of GDDR5X memory and 3584 CUDA cores).

4.4. Experimental results

4.4.1. Linguistic information extraction results

We perform linguistic information extraction experiments on the selected six regulatory rules using the developed automated regulatory rules processing system with the following steps:

(1) Firstly, the dependency trees, with both POS tags and dependency labels, are created as the outputs of the pre-processing and feature generation stages. With this step, each regulatory rule sentence is properly decomposed into minimal units (tokens) with linguistic properties and inter-unit semantic relations (dependency) are generated for further analysis. As an example, the output dependency tree of the regulatory rule "Wear a hard hat on a construction site" is shown in Fig. 15.

(2) Subsequently, semantic analysis is performed by ontology modeling to automatically extract the entities from the dependency trees generated in the previous step, together with their attributes and relations between them. Up to this point, each regulatory rule has been automatically transformed from a natural language sentence into a model composed of multiple inter-related instances capable of describing scenes. Fig. 16 demonstrates an example of the ontology modeling of the regulatory rule "Wear a hard hat on a construction site".

(3) Next, the information extracted from the ontology model is encoded in logic representations using semantic phrases and logical connectives in order to further structure the linguistic information in regulatory rules. The logic representations of the selected six regulatory rules are shown in Table 9 which correctly demonstrated the extracted entities and requirements from regulatory rules.

(4) Finally, based on the logic representation, the hierarchical graph $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$ is generated for linguistic information representation as demonstrated in Fig. 17, where $\widehat{V}=\{$*person*, *construction site*, *height*, *grinder*, *hard hat*, *body harness*, *dust mask*, *safety glasses*, *face*, *two hands*$\}$, the edges $\widehat{E}=\{$(*person*, *construction site*, *on*), (*person*, *height*, *on*), (*person*, *grinder*, *operate*), (*person*, *hard hat*, *wear*, *obligation*), (*person*, *body harness*, *use*, *obligation*), (*person*, *dust mask*, *wear*, *obligation*), (*person*, *safety glasses*, *wear*, *obligation*), (*person*, *face*, *near*, *prohibition*), (*person*, *use*, *two hands*, *obligation*), (*two hands*, *operate*, *grinder*)$\}$, and the conditional relations set $\widehat{C}=\{$((*person*, *construction site*, *on*))⇀((*person*, *hard hat*, *wear*, *obligation*), (*person*, *dust mask*, *wear*, *obligation*)), ((*person*, *height*, *on*)) ⇀ ((*person*, *body harness*, *use*, *obligation*)), ((*person*, *grinder*, *operate*)) ⇀ ((*person*, *safety glasses*, *wear*, *obligation*), (*person*, *face*, *near*, *prohibition*), (*person*, *use*, *two hands*, *obligation*), (*two hands*, *operate*, *grinder*))$\}$. $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$ contains all linguistic information of the target regulatory rules which demonstrated the effective of

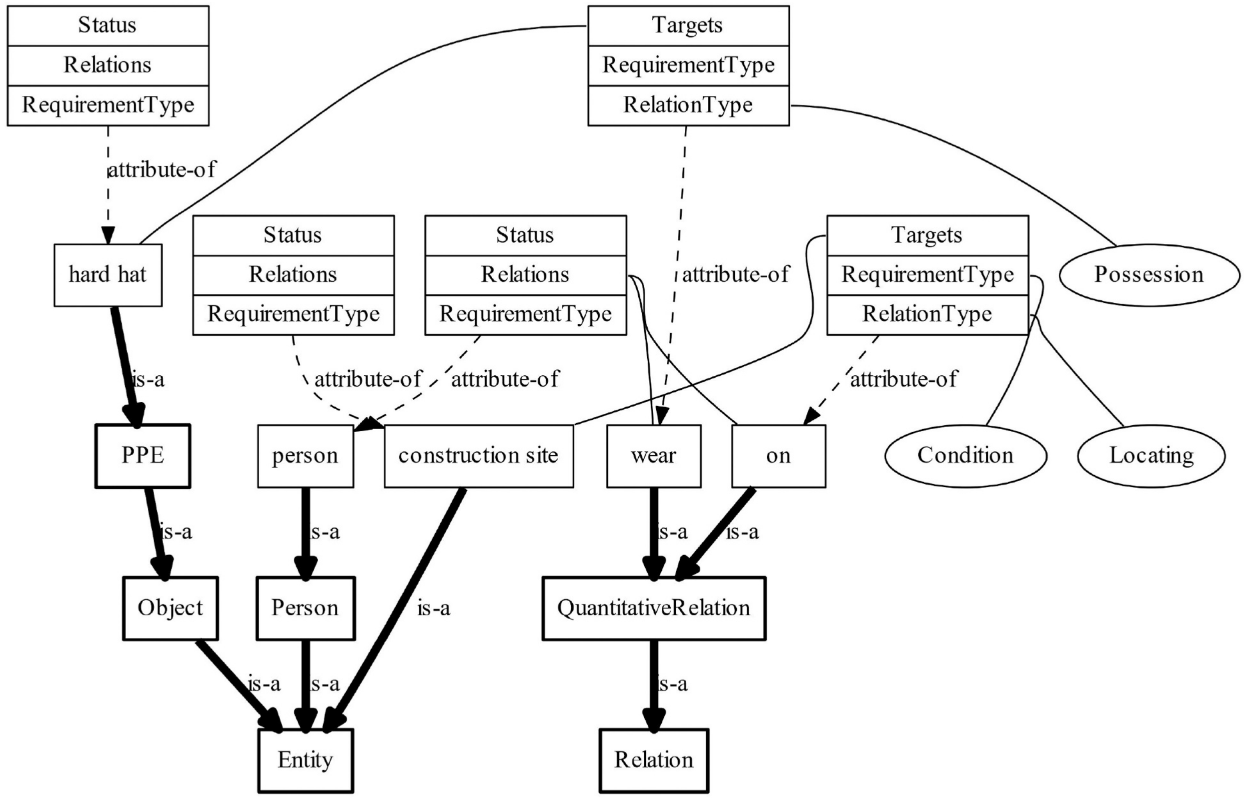


Fig. 16. Ontology modeling for regulatory rule: "Wear a hard hat on a construction site."

Table 9
Logic representation of regulatory rules from on-site occupational safety guides.

No.	Regulatory rules	Logic representation
1	Wear a hard hat on a construction site.	$((person, on, construction\ site)) \rightarrow () \wedge ((person, wear, hard\ hat))$
2	Wear a dust mask on a construction site.	$((person, on, construction\ site)) \rightarrow () \wedge (person, wear, dust\ mask))$
3	Use body harness when working on height.	$((person, on, height)) \rightarrow () \wedge ((person, use, body\ harness))$
4	Wear a safety glasses when operating a grinder.	$((person, operate, grinder)) \rightarrow () \wedge ((person, wear, safety\ glasses))$
5	Always use two hands when operating a grinder.	$((person, operate, grinder)) \rightarrow () \wedge ((person, use, two\ hands) \wedge (two\ hands, operate, grinder))$
6	Never operate a grinder near face.	$((person, operate, grinder)) \rightarrow (\neg(grinder, near, face)) \wedge ()$

the proposed approach on automated linguistic information extraction and representation.

The automated generated hierarchical graph $\widehat{G}(\widehat{V}, \widehat{E}, \widehat{C})$ was reviewed by a team of researchers and on-site managers, and the correct representation of the six target regulatory rules was finally confirmed. Fig. 18 visualizes the subgraphs of each rule included in $\widehat{G}(\widehat{V}, \widehat{E}, \widehat{C})$, from which it is observed that each of the six regulatory rules is correctly represented by the graph structure. In addition, we provide more results of linguistic information extraction for different scenes in Appendix A.

4.4.2. Hazards identification results

We evaluate the performance of the proposed model on our testing dataset to address compliance with the six regulatory rules. For each on-site image the entities in it are first detected and a scene graph $G(V,E)$ is automatically generated to represent its scene information. When a hazard identification is performed, the subgraph composed of relevant regulatory rules are automatically extracted first to specify the reasoning targets and reduce the computational complexity. As the testing dataset contains two scenes, i.e."working on height" and "operating a grinder", two relevant rules subgraphs $\widehat{G}_h(\widehat{V}_h, \widehat{E}_h)$ and $\widehat{G}_g(\widehat{V}_g, \widehat{E}_g)$ are automatically generated as illustrated in Fig. 19(b) and Fig. 19(c), respectively. Reasoning for hazards identification is performed based on compliance checking between the $G(V,E)$ and its relevant rules subgraph. Fig. 20 illustrates the identification examples on the testing dataset. Our model detects the individuals and objects in the obtained image (Fig. 20(a)) and visualizes an intuitive scene graph in Fig. 20(b). The hazards identification results are shown in Fig. 20(c).

The quantitative results under different scenes are reported in Table 10 and Table 11. The resolution of the individual regions in the testing image gradually became smaller as the distance between the camera and individuals increased. However, benefiting from YOLOv4's ability of detecting small objects, the identification precision and recall of the proposed model are not significantly affected. For the scene "working on height", the proposed model successfully identifies improper hard hat use in testing images at all distances with 100% overall precision and 99.89% overall recall. And for the identification results of proper dust mask use, the proposed model is also able to achieve significant performance with with 100% overall precision and 99.74% overall recall. Furthermore, in the experiments of proper body harness use identification, the performance of the proposed model is slightly decreased on the side view in far-field images (92.97% recall in the 7 m case), but still achieves 100% overall precision and 97.01% overall recall. For the scene "operating a grinder", most of the PPE (safety glasses and dust masks) in the testing images are relatively small due to the operating postures and camera angles. The grinder is also not easily identifiable due to the angle and hand obscuration. All of these add to the difficulty of the model for hazards identification. For the identification results of proper safety glasses use, which is considered to be the most challenging in the experiments because most safety glasses in the testing images are transparent and the recognition will be

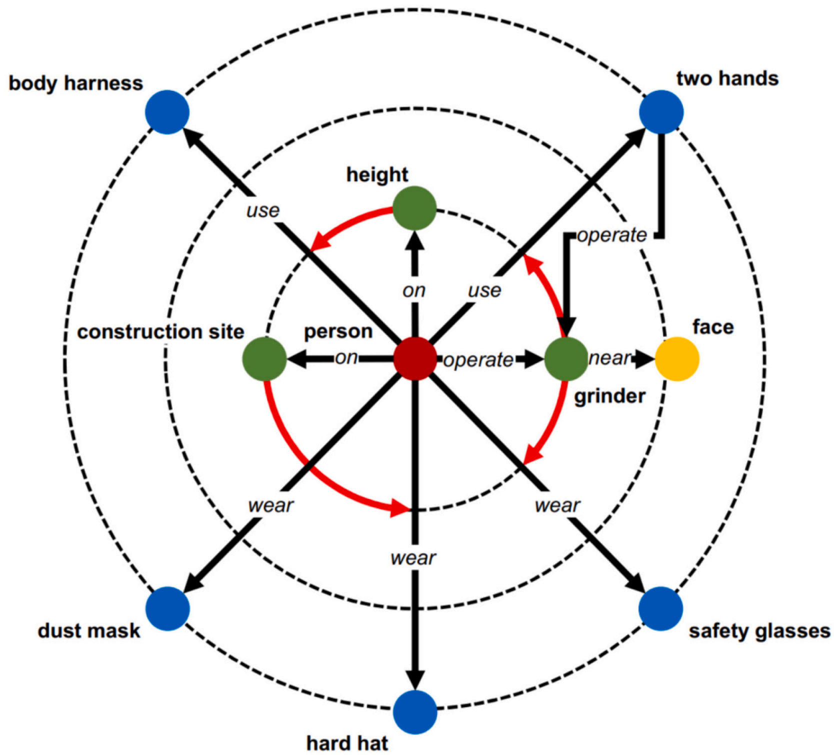


Fig. 17. The hierarchical graph generated from on-site occupational safety guides.

seriously affected by the level of the illumination, the proposed model achieves 96.40% precision and 70.76% recall. The proposed model also achieves notable performance at 99.90% precision and 95.62% recall for the identification results of proper dust mask in scene "operating a grinder" despite the relatively small dust masks in the testing images. Additionally, the identification of hazardous operation of the grinder is also well accomplished by the proposed model. For the hazardous behavior of single-handed operation of a grinder, the proposed model is also able to achieve significant performance with 97.01% precision and 100% recall. The dangerous behavior of operating a grinder near face is also successfully identified with 98.60% precision and 88.49% recall.

To benchmark the performance of our proposed model, We also perform the hazards identification experiments on the YOLOv3-based model introduced in our previous work [24], which has achieved state-of-the-art performance for improper PPE use identification. This result serves as the baseline for the following comparisons. As listed in Table 10 and Table 11, the proposed model in this paper achieves better performance than the baseline model in both "working on height" and "operating a grinder" scenes. Notably, the recall rate of proper dust mask identification and near-face grinder operating identification in the experiments of the "operating a grinder" scene improved significantly compared to the baseline model (95.62% vs. 70.99%, 88.49% vs. 79.23%). Overall, the proposed model outperforms the baseline model in terms of both precision ("working on height" scene: 100% vs. 99.99%, "operating a grinder" scene: 98.60% vs. 97.37%) and recall ("working on height" scene: 99.02% vs. 97.78%, "operating a grinder" scene: 88.49% vs. 79.23%).

5. Discussion

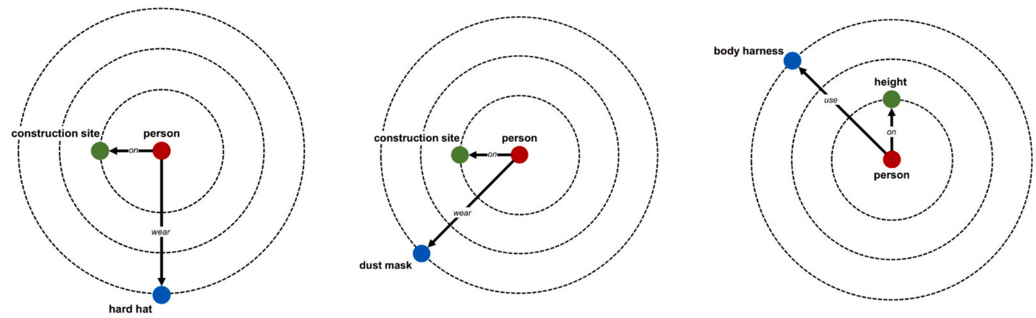
Construction workers are required to follow on-site regulatory rules in order to avoid occupational hazards. However, compliance of regulatory rules is not strictly enforced among workers due to all kinds of reasons. Even though vision-based on-site hazards identification systems have been widely used to enhance compliance checking, it remains to be suggested that on-site hazards identification systems with the ability to automatically understand and dynamically adapt to changes in regulatory rules to improve practicality, as regulatory rules vary from region to region and may change due to external conditions.

5.1. Contributions

To address current gaps noted above, this paper proposes a graph-based framework that integrates linguistic and visual information to process regulatory rule sentences and images for on-site occupational hazards identification. Particularly, the following originalities and achievements are thought to be the contribution of this paper.

The linguistic information extraction approach proposed in this paper originally deploys NLP-based syntactic structure analysis with the ontology concept to automatically extract and represent key information from regulatory rules in AEC domains. The extracted linguistic information is represented by a novel hierarchical graph structure introduced in our previous work [24], which is further refined in this paper by addressing the prohibition relationships in regulatory rules. In contrast to other integrated models [18–23], our model is capable of automatically extracting and representing the linguistic information of regulatory rules and using a unified model for hazards reasoning without the requirement of third-party software. As for on-site deployment, our system allows direct reading of textual data (e.g., regulatory documents, meeting memos) for automated processing, and safety monitoring of different regions of the site can be performed without any system setup by site managers, which is more practical than the other integrated models surveyed. As demonstrated in Fig. 18, the key information contained in the six target regulatory rules that require addressing in the current experiment are all correctly represented in the graph structure. In addition, in the hazards reasoning experiments, the proposed approach properly extracts the corresponding subgraphs of relevant regulatory rules from the generated hierarchical graph in response to the visual information of different scenes (Fig. 19).

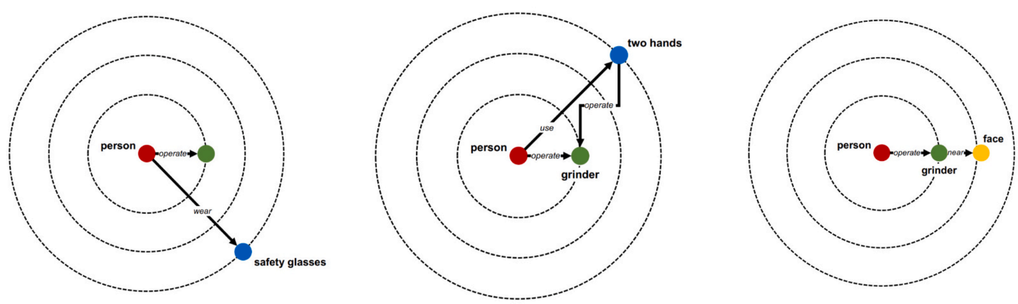
A novel scene understanding approach is presented in this paper



(a) The subgraph of regulatory rule: "Wear a hard hat on a construction site."

(b) The subgraph of regulatory rule: "Wear a dust mask on a construction site."

(c) The subgraph of regulatory rule: "Use body harness when working on height."

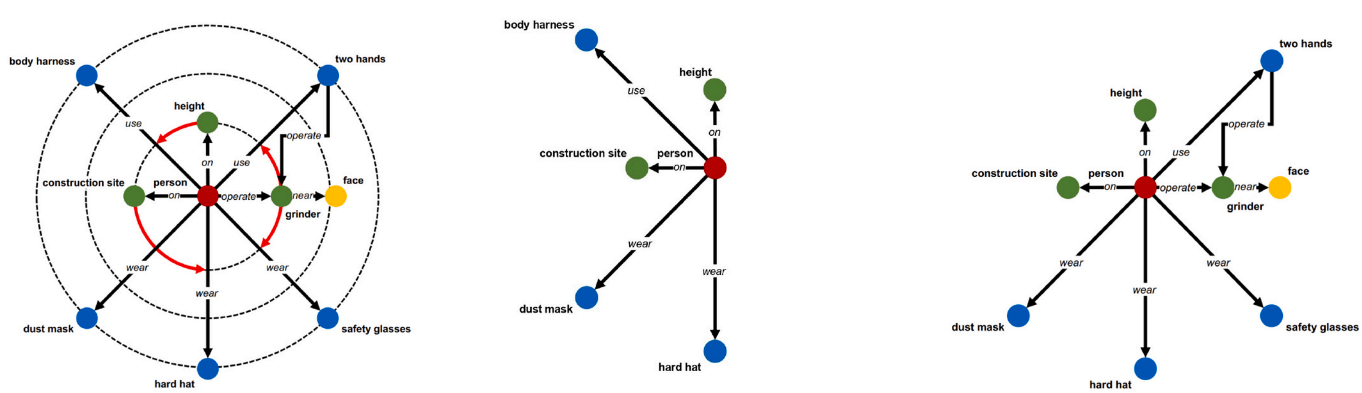


(d) The subgraph of regulatory rule: "Wear a safety glasses when operating a grinder."

(e) The subgraph of regulatory rule: "Always use two hands when operating a grinder."

(f) The subgraph of regulatory rule: "Never operate a grinder near face."

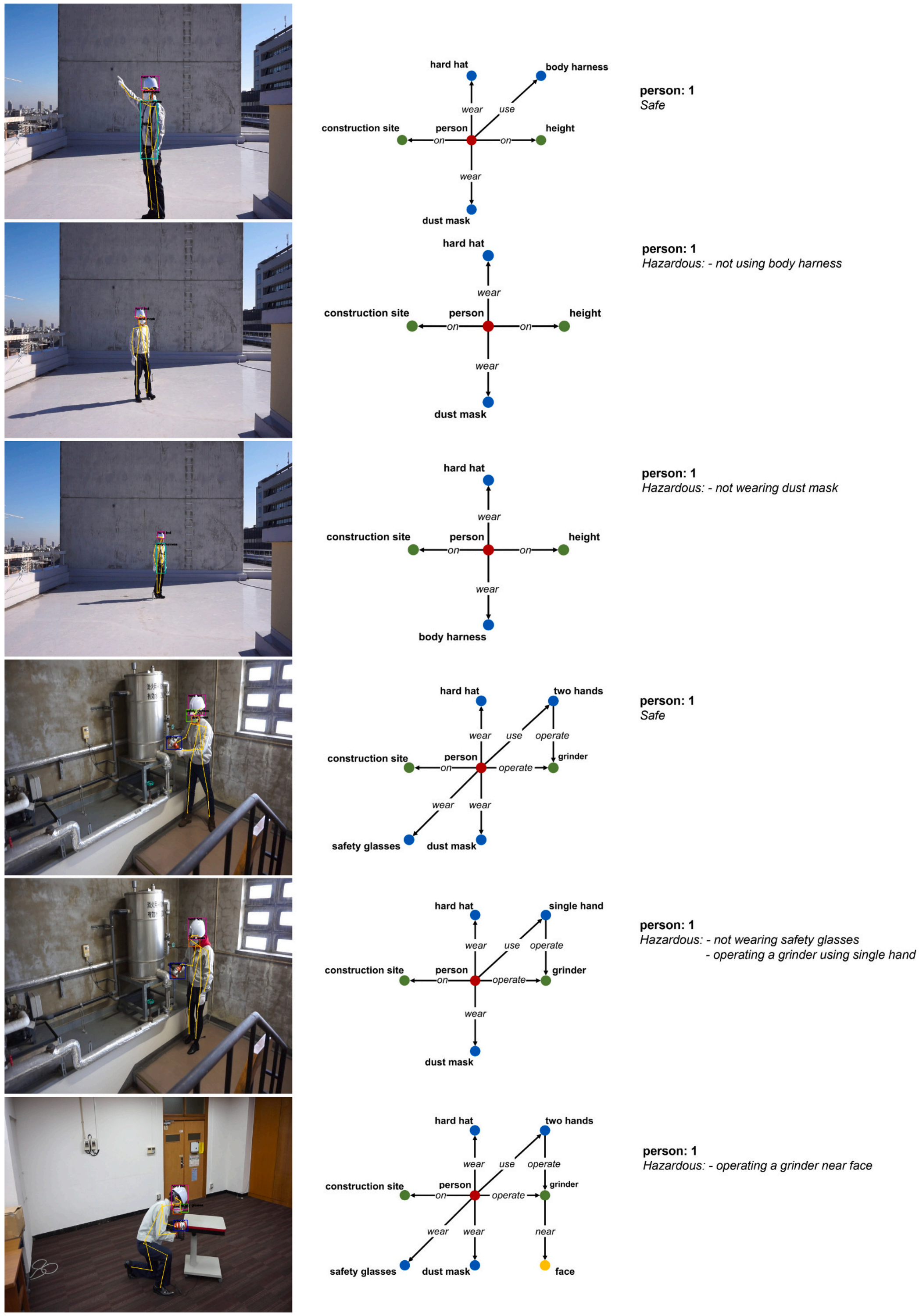
Fig. 18. Subgraph visualization of the regulatory rules included in $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$.



(a) (b) (c)

Fig. 19. (b)$\widehat{G}_h(\widehat{V}_h,\widehat{E}_h)$ and (c) $\widehat{G}_g(\widehat{V}_g,\widehat{E}_g)$ are the relevant rules graphs extracted from (b) regulatory rules hierarchical graph $\widehat{G}(\widehat{V},\widehat{E},\widehat{C})$ and contains all regulatory rules for the situation of target scenes"working on height" and "operating a grinder".

employing scene graph as the basic notion of information representation structure to automatically perceive visual information from images. The extraction of entities in the scene combines deep learning-based object detection and pose estimation as human skeletons provide more fine-grained information about a person, especially in the case of occlusion, and is more effective for multi-hazard identification task, which has been verified in our previous work [24] to have better performance compared to the commonly used object detection-based approaches in



(a) Entity detection results (b) Scene graph representation (c) Hazards identification results

Fig. 20. Identification examples on the testing dataset.

Table 10
Hazards identification results of the scene "working on height".

Distance (m)	Model	Hard hat		Dust mask		Body harness	
		Precision (%)	Recall (%)	Precision (%)	Recall (%)	Precision (%)	Recall (%)
3	Baseline	100	95.50	100	100	100	100
	This paper	**100**	**100**	**100**	**100**	**100**	**100**
5	Baseline	100	94.78	99.87	99.73	100	98.27
	This paper	**100**	**100**	**100**	**99.87**	**100**	**98.40**
7	Baseline	100	90.89	100	95.74	100	92.28
	This paper	**100**	**99.89**	**100**	**99.35**	**100**	**92.97**
Overall	Baseline	100	96.74	99.96	99.70	100	96.97
	This paper	**100**	**99.96**	**100**	**99.74**	**100**	**97.01**

Table 11
Hazards identification results of the scene "operating a grinder".

Model	Hard hat		Safety glasses		Dust mask		Grinder[1]		Grinder[2]	
	Precision (%)	Recall (%)	Precision (%)	Recall (%)	Precision (%)	Recall (%)	Precision (%)	Recall (%)	Precision (%)	Recall (%)
Baseline	99.19	99.59	93.17	69.98	99.74	70.99	95.09	100	97.37	79.23
This paper	**99.32**	**100**	**96.40**	**70.76**	**99.90**	**95.62**	**97.01**	**100**	**98.60**	**88.49**

[1] operate a grinder using single hand.
[2] operate a grinder near face.

the surveyed vision-based works [11–15]. According to our previous work [24], this paper presents some improvements in visual information extraction. We employ the existing state-of-the-art vision-based object detection model, i.e., YOLOv4, which is one of the most performing novel object detection models currently available compared with other object detection models used in [24] and other surveyed works [11–17].

Additionally, apart from the proper use identification of PPE addressed in our previous work [24], the proposed automated reasoning module in this paper also performs the identification of the safe use of tools, which is a more complex task. Effectiveness of the model proposed in this paper were experimentally evaluated on two scenes, i.e., "working on height" and "operating a grinder". To benchmark the performance, results were compared with the model introduced in our previous work [24], and reported in Table 10 and Table 11. The results demonstrate that the model proposed in this paper achieves better perceptual capabilities for scene understanding, especially in the complex scenes of safe operation of grinder.

5.2. Limitations

Further extensions of this work will be investigated in the following directions considering the limitations of the approach proposed in this paper.

The linguistic information extraction approach proposed in this paper utilizes a pre-trained spaCy parser to generate syntactic features, however we observe that the parser fails in generating correct POS tags and dependency labels when addressing some uncommon construction-specific words, which would result in the subsequent semantic analysis not proceeding properly. In our coming work, we plan to collect and annotate extensive construction-related documents to train a better parser to solve this issue.

Although the model proposed in this paper obtains a better performance with respect to the baseline model [24] for automated identification of the safe operation of grinder, we nevertheless notice the inadequacy of the 2D geometric distance in individual-object relationship analysis in the examples of failed identification. To address this gap, future enhancements will consider the use of 3D pose estimation and object detection models.

The hazards reasoning approach presented in this paper is based on frame-by-frame graph structure analysis without considering time-series relationships. However, a hazard behavior is usually composed of multiple hazard actions, and the introduction of time-series analysis provides a more precise description of hazard behaviors. In our future work, we will investigate the use of time-series analysis models to dynamize the graph structure for hazards identification and prediction.

6. Conclusion

This paper has presented a novel approach integrating linguistic and visual information to process regulatory rule sentences and images for on-site occupational hazards identification. First, we propose a regulatory rules processing approach to automatically extract and represent the key linguistic information of regulatory rules that is intended to indicate the potential occupational hazards which need to be identified. Subsequently, we introduce a vision-based image scene information understanding approach to process on-site images by the combination of deep learning-based object detection and individual detection using geometric relationships analysis. Additionally, an automated reasoning approach is proposed to provide the integration of the processed linguistic and visual information and perform hazards identification. The hazards of two scenes, i.e., "working on height" and "operating a grinder", were successfully identified. The test results indicate that our model was capable of identifying the on-site occupational hazards with high precision ("working on height" scene: 100%, "operating a grinder" scene: 98.60%) and recall rate ("working on height" scene: 99.02%, "operating a grinder" scene: 88.49%) under different environmental conditions. Further improvements will be to train a better syntactic parser for ensuring the proper extraction of linguistic information, and to improve the performance of visual information representation by introducing 3D entity extraction and time-series analysis.

Declaration of Competing Interest

None.

Acknowledgments

This research was funded by Japan Society for the Promotion of Science (JSPS) Grants-in-Aid for Scientific Research Grant No. 19K05324.

Appendix A. Additional linguistic information extraction results

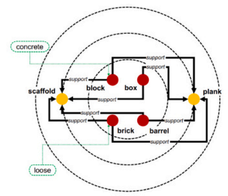
To further validate the correctness and generalizability of the developed automated regulation processing system, we performed linguistic information extraction experiments on regulatory rules from the OSHA Pocket Guide "Worker Safety Series: Construction" [3], with the following selected regulatory rules shown below, which contain different construction scenes (e.g. scaffolding, forklifts, etc.):

(1) "Unstable objects, such as barrels, boxes, loose bricks or concrete blocks must not be used to support scaffolds or planks."
(2) "Scaffold must be equipped with guardrails, midrails and toeboards."
(3) "Scaffolds must be at least 10 feet from electric power lines at all times."
(4) "(Worker on height) Use safety net systems or personal fall arrest systems."
(5) "Avoid using ladders with metallic components near electrical work and overhead power lines."
(6) "Never enter an unprotected trench."
(7) "(Forklift) do not move a load over workers."
(8) "Operators shall always wear seatbelts."
(9) "Construction workers should wear work shoes or boots with slip-resistant and puncture-resistant soles."
(10) "Do not raise or lower the forks while the forklift is moving."

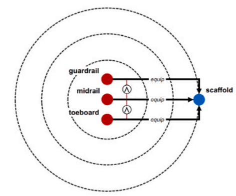
The logic representations of the selected regulatory rules are shown in Table A.12 which correctly demonstrated the extracted entities and requirements from regulatory rules. Additionally, Fig. A.21 visualizes the generated graphs of the selected regulatory rules. Although these regulatory rules cover different scenes at construction sites and are relatively more complex in terms of syntactic structure, the developed automated regulation processing system still correctly represents their linguistic information in graph structure.

Table A.12
Logic representation of the regulatory rules from the OSHA Pocket Guide.

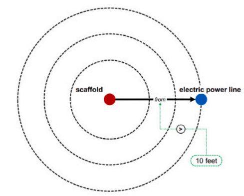
No.	Regulatory rules	Logic representation
1	Unstable objects, such as barrels, boxes, loose bricks or concrete blocks must not be used to support scaffolds or planks.	$(\neg(barrel, support, scaffold) \vee \neg(box, support, scaffold) \vee \neg(brick(be, loose), support, scaffold) \vee \neg(block(be, concrete), support, scaffold) \vee \neg(barrel, support, plank) \vee \neg(box, support, plank) \vee \neg(brick(be, loose), support, plank) \vee \neg(block(be, concrete), support, plank)) \wedge ()$
2	Scaffold must be equipped with guardrails, midrails and toeboards.	$() \wedge ((guardrail, equip, scaffold) \wedge (midrail, equip, scaffold) \wedge (toeboard, equip, scaffold))$
3	Scaffolds must be at least 10 ft from electric power lines at all times.	$() \wedge ((scaffold, from(>10\ feet), electric\ power\ line))$
4	(Worker on height) Use safety net systems or personal fall arrest systems.	$((worker, on, height)) \rightarrow () \wedge ((worker, use, safety\ net\ system) \vee (worker, use, personal\ fall\ arrest\ system))$
5	Avoid using ladders with metallic components near electrical work and overhead power lines.	$((worker, use, ladders(be, metallic\ components)) \rightarrow (\neg(ladders(be, metallic\ components), near, electricalwork) \wedge \neg(ladders(be, metallic\ components), overhead, powerlines)) \wedge ()$
6	Never enter an unprotected trench.	$(\neg(worker, enter, trench(be, unprotected)) \wedge ()$
7	(Forklift) do not move a load over workers.	$((forklift, move, load)) \rightarrow (\neg(load, over, worker)) \wedge ()$
8	Operators shall always wear seatbelts.	$() \wedge ((operator, wear, seatbelt))$
9	Construction workers should wear work shoes or boots with slip-resistant and puncture-resistant soles.	$() \wedge ((construction\ worker, wear, work\ shoes) \vee (construction\ worker, wear, boots(be, (slip-resistant \wedge puncture-resistant\ soles)))$
10	Do not raise or lower the forks while the forklift is moving.	$((forklift, be, moving)) \rightarrow (\neg(worker, raise, forks) \wedge \neg(worker, lower, forks)) \vee ()$



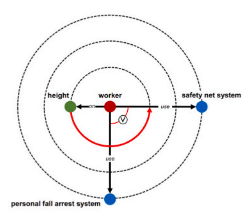
(a) "Unstable objects, such as barrels, boxes, loose bricks or concrete blocks must not be used to support scaffolds or planks."



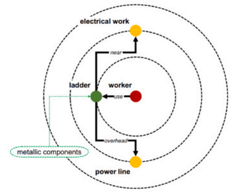
(b) "Scaffold must be equipped with guardrails, midrails and toeboards."



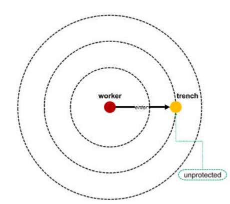
(c) "Scaffolds must be at least 10 feet from electric power lines at all times."



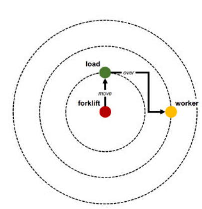
(d) "(Worker on height) Use safety net systems or personal fall arrest systems."



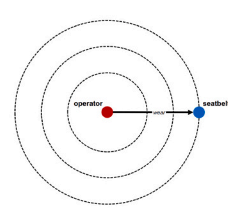
(e) "Avoid using ladders with metallic components near electrical work and overhead power lines."



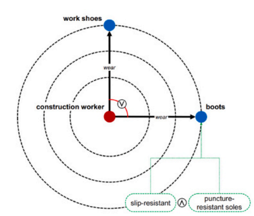
(f) "Never enter an unprotected trench."



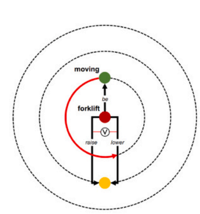
(g) "(Forklift) do not move a load over workers."



(h) "Operators shall always wear seatbelts."



(i) "Construction workers should wear work shoes or boots with slip-resistant and puncture-resistant soles."

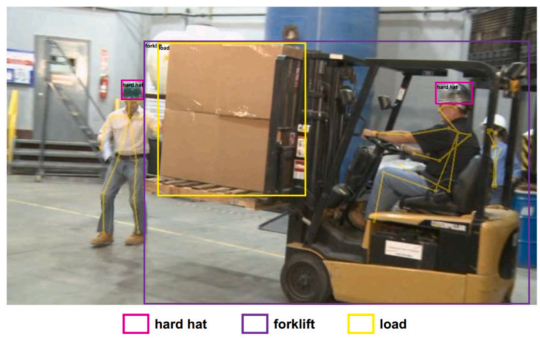


(j) "Do not raise or lower the forks while the forklift is moving."

Fig. A.21. The hierarchical graphs generated from the OSHA Pocket Guide.

Appendix B. Further discussion on extendability in different scenes

According to the above regulatory rule representations, the task of hazards identification can be extended to other scenes by adding the corresponding categories (e.g. ladders, forklifts, etc.) of object detection in the graph structure together with geometric feature analysis. Fig. B.22 provides some examples of how the proposed approach can be extended to other scenes with individuals detected using OpenPose and objects that can be detected in the future by training a YOLOv4 model containing more categories. For example, for the regulatory rule "(Forklift) do not move a load over workers.", the hazard of a scene can be reasoned by analyzing the geometric position of the detected individual with respect to the detected load of a forklift (whether it is overhead) as shown in Fig. B.22(a). And for the regulatory rule "Operators shall always wear seatbelts.", as demonstrated in Fig. B.22(b) the hazard of whether or not wearing a seatbelt can be reasoned through similar analysis as in section 3.3.2 regarding the relationship between individuals and body harnesses. In addition for the regulatory rule "Construction workers should wear work shoes or boots with slip-resistant and puncture-resistant soles.", similar to the analysis of the relationship between individuals and hard hats in section 3.3.2, it is possible to identify whether work shoes are worn by analyzing the geometric position of a detected individual's ankles (body parts 10 and 13 in Fig. 6) in relation to the detected work shoes as shown in Fig. B.22 (c).



(a) "(Forklift) do not move a load over workers."

(b) "Operators shall always wear seatbelts."

(c) "Construction workers should wear work shoes or boots with slip-resistant and puncture-resistant soles."

Fig. B.22. Examples of how the proposed approach extends to other scenes.

References

[1] Bureau of Labor Statistics, Industries at a Glance: Construction. https://www.bls.gov/iag/tgs/iag23.htm, 2021. Accessed date: 9 February 2021.

[2] Japan Industrial Safety and Health Association, OSH Statistics in Japan, https://www.jisha.or.jp/english/statistics/index.html, Accessed date: 9 February 2021.

[3] Occupational Safety & Health Administration, Worker safety series: construction, https://www.osha.gov/Publications/OSHA3252/3252.html, Accessed date: 9 January 2022.

[4] S. Konda, H.M. Tiesman, A.A. Reichard, Fatal traumatic brain injuries in the construction industry, 2003-2010, Am. J. Ind. Med. 59 (2016) 212–220, https://doi.org/10.1002/ajim.22557.

[5] Occupational Safety & Health Administration, Safety and Health Regulations for Construction: Head protection, https://www.osha.gov/laws-regs/regulations/standardnumber/1926/1926.100, Accessed date: 9 February 2021.

[6] National Institute for Occupational Safety and Health, EYE SAFETY, https://www.cdc.gov/niosh/topics/eye/, Accessed date: 9 February 2021.

[7] A.L. Dannenberg, L.M. Parver, R.J. Brechner, L. Khoo, Penetration eye injuries in the workplace: the National eye Trauma System Registry, Arch. Ophthalmol. 110 (6) (1992) 843–848, https://doi.org/10.1001/archopht.1992.01080180115038.

[8] Government of Western Australia, Department of Commerce, WorkSafe Division, Guide to using dust masks in construction work, https://www.commerce.wa.gov.au/sites/default/files/atoms/files/guide_to_using_dust_mask.pdf, Accessed date: 9 February 2021.

[9] Occupational Safety & Health Administration, Occupational Safety and Health Standards: Eye and face protection, https://www.osha.gov/laws-regs/regulations/standardnumber/1910/1910.133, Accessed date: 9 February 2021.

[10] Bureau of Labor Statistics, Census of Fatal Occupational Injuries Summary. https://www.bls.gov/news.release/cfoi.nr0.htm, 2018. Accessed date: 9 February 2021.

[11] Q. Fang, H. Li, X. Luo, L. Ding, H. Luo, T.M. Rose, W. An, Detecting non-hardhat-use by a deep learning method from far-field surveillance videos, Autom. Constr. 85 (2018) 1–9, https://doi.org/10.1016/j.autcon.2017.09.018.

[12] J. Wu, N. Cai, W. Chen, H. Wang, G. Wang, Automatic detection of hardhats worn by construction personnel: a deep learning approach and benchmark dataset, Autom. Constr. 106 (2019), 102894, https://doi.org/10.1016/j.autcon.2019.102894.

[13] N.D. Nath, A.H. Behzadan, S.G. Paal, Deep learning for site safety: real-time detection of personal protective equipment, Autom. Constr. 112 (2020), 103085, https://doi.org/10.1016/j.autcon.2020.103085.

[14] N.D. Nath, A.H. Behzadan, Deep convolutional networks for construction object detection under different visual conditions, Front. Built Environ. 6 (2020) 97, https://doi.org/10.3389/fbuil.2020.00097.

[15] J. Shen, X. Xiong, Y. Li, W. He, P. Li, X. Zheng, Detecting safety helmet wearing on construction sites with bounding-box regression and deep transfer learning, Comp. Aided Civil Infrastruct. Eng. (2020) 1–17, https://doi.org/10.1111/mice.12579.

[16] S. Chen, K. Demachi, A vision-based approach for ensuring proper use of personal protective equipment (PPE) in decommissioning of Fukushima daiichi nuclear power station, Appl. Sci. 10 (2020) 5129, https://doi.org/10.3390/app10155129.

[17] R. Xiong, P. Tang, Pose guided anchoring for detecting proper use of personal protective equipment, Autom. Constr. 130 (2021), 103828, https://doi.org/10.1016/j.autcon.2021.103828.

[18] R. Xiong, Y. Song, H. Li, Y. Wang, Onsite video mining for construction hazards identification with visual relationships, Adv. Eng. Inform. 42 (2019), 100966, https://doi.org/10.1016/j.aei.2019.100966.

[19] M. Zhang, M. Zhu, X. Zhao, Recognition of high-risk scenarios in building construction based on image semantics, J. Comput. Civ. Eng. 34 (2020) 04020019, https://doi.org/10.1061/(asce)cp.1943-5487.0000900.

[20] W. Fang, L. Ma, P.E.D. Love, H. Luo, L. Ding, A. Zhou, Knowledge graph for identifying hazards on construction sites: integrating computer vision with ontology, Autom. Constr. 119 (2020), 103310, https://doi.org/10.1016/j.autcon.2020.103310.

[21] H. Wu, B. Zhong, H. Li, P. Love, X. Pan, N. Zhao, Combining computer vision with semantic reasoning for on-site safety management in construction, J. Build. Eng. 42 (2021) 103036, https://doi.org/10.1016/j.jobe.2021.1030360.

[22] N. Khan, M.R. Saleem, D. Lee, M.W. Park, C. Park, Utilizing safety rule correlation for mobile scaffolds monitoring leveraging deep convolution neural networks, Comput. Ind. 129 (2021), 103448, https://doi.org/10.1016/j.compind.2021.103448.

[23] H. Liu, G. Wang, T. Huang, P. He, M. Skitmore, X. Luo, Manifesting construction activity scenes via image captioning, Autom. Constr. 119 (2020), 103334, https://doi.org/10.1016/j.autcon.2020.103334.

[24] S. Chen, K. Demachi, Automation in construction towards on-site hazards identification of improper use of personal protective equipment using deep learning-based geometric relationships and hierarchical scene graph, Autom. Constr. 125 (2021) 103619, https://doi.org/10.1016/j.autcon.2021.103619.

[25] G. Grefenstette, P. Tapanainen, What is a word, what is a sentence? Problems of tokenization, in: Proceedings of the 3rd International Conference on Computational Lexicography, 1994, pp. 79–87. https://citeseerx.ist.psu.edu/viewdoc/download?doi=10.1.1.31.8947rep=rep1type=pdf.

[26] J. Miller, A Critical Introduction to Syntax, Continuum International Publishing Group, 2011. ISBN: 9780826497048.

[27] Z. Cao, G. Hidalgo Martinez, T. Simon, S.-E. Wei, Y.A. Sheikh, OpenPose: Realtime multi-person 2D pose estimation using part affinity fields, IEEE Trans. IEEE Trans. Pattern Anal. Machine Intel. 43 (2021) 172–186, https://doi.org/10.1109/tpami.2019.2929257.

[28] T. Lin, M. Maire, S. Belongie, L. Bourdev, R. Girshick, J. Hays, P. Perona, D. Ramanan, C.L. Zitnick, P. Dollár, Microsoft COCO: common objects in context, Proc. Eur. Conf. Comp. Vision (2014) 740–755, https://doi.org/10.1007/978-3-319-10602-1_48.

[29] C.Y. Wang, A. Bochkovskiy, H.Y.M. Liao, Scaled-yolov4: scaling cross stage partial network, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2021) 13029–13038. https://openaccess.thecvf.com/content/CVPR2021/html/Wang_Scaled-YOLOv4_Scaling_Cross_Stage_Partial_Network_CVPR_2021_paper.html.

[30] J. Redmon, S. Divvala, R. Girshick, A. Farhadi, You only look once: unified, real-time object detection, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2016) 779–788, https://doi.org/10.1109/CVPR.2016.91.

[31] J. Redmon, A. Farhadi, YOLO9000: Better, faster, stronger, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2017) 6517–6525, https://doi.org/10.1109/CVPR.2017.690.

[32] J. Redmon, A. Farhadi, YOLOv3: An incremental improvement, arXiv preprint, http://arxiv.org/abs/1804.02767.

[33] C.Y. Wang, H.Y.M. Liao, I.H. Yeh, Y.H. Wu, P.Y. Chen, J.W. Hsieh, CSPNET: a new backbone that can enhance learning capability of CNN, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2020) 1571–1580, https://doi.org/10.1109/CVPRW50498.2020.00203.

[34] K. He, X. Zhang, S. Ren, J. Sun, Spatial pyramid pooling in deep convolutional networks for visual recognition, IEEE Trans. Pattern Anal. Mach. Intell. 37 (2015) 1904–1916, https://doi.org/10.1109/TPAMI.2015.2389824.

[35] T.-Y. Lin, P. Dollár, R. Girshick, K. He, B. Hariharan, S. Belongie, Feature pyramid networks for object detection, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2017) 2117–2125, https://doi.org/10.1109/CVPR.2017.106.

[36] S. Liu, L. Qi, H. Qin, J. Shi, J. Jia, Path aggregation network for instance segmentation, Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (2018) 8759–8768, https://doi.org/10.1109/CVPR.2018.00913.

[37] S.M. Hsieh, C.C. Hsu, L.F. Hsu, Efficient method to perform isomorphism testing of labeled graphs, Proc. Int. Conf. Comp. Sci. Appl. (2006) 422–431, https://doi.org/10.1007/11751649_46.

[38] spaCy: Industrial-strength NLP. https://spacy.io/. Accessed date: 9 February 2021.

[39] NetworkX: Network Analysis in Python. https://networkx.org/. Accessed date: 9 February 2021.

[40] J. Ellson, E. Gansner, L. Koutsofios, S.C. North, G. Woodhull, Graphviz - open source graph drawing tools, Proc. Int. Symp. Graph Draw. (2001) 483–484, https://doi.org/10.1007/3-540-45848-4_57.